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Public Transport Launch Center Management System

Senior project 1

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Chapter One

Introduction and Project Definition

Introduction to the Project Topic

1 Overview:

The starting point will be public transportation in the Launch Center system: the project will begin with the city of Damascus and Launch Centers (public garages), then expand to all cities and governorates of Syria. The Launch Centers.

The project aims to design and develop an integrated digital system for managing public transportation, including scheduling, handling public transportation operations, organizing trips, tracking buses, addressing passenger complaints, and providing analytical reports.

2-Project Justifications

To address organizational problems due to reliance on administrative procedures following a hierarchical sequence, using manual mail.

To reduce manipulation of fare prices and prevent exploitation.

Increasing reliance on public transportation requires smart systems that help improve the user experience, reduce congestion, and enhance operational

oversight. The project addresses issues resulting from manual management, such as delayed trips and poor resource distribution.

3-Project Objective

To create an integrated information system for the Launch Center that facilitates daily management operations, and provides senior management with data-supported tools for analysis and decision-making.

Developing an electronic platform that facilitates travel between cities

Improving the reality of public transportation services

4-Project Scope Assumptions and Restrictions

The project deals with managing the operational aspects of the transport institution by developing a system that includes managing vehicles, drivers, routes, schedules, and performance monitoring

4-1-Assumptions

4-2- Restrictions

5- Research Methodology

. The project follows an analytical approach in studying requirements and designing the system .

6- Organizational Document Structure –

Report Structure and Objectives

The report is organized into five main chapters covering all aspects of the project:

⌘ Chapter One: Introduction and Project Scope

This chapter provides the general framework of the project, outlining its objectives, which consist of establishing a public transport hub management system and defining the project scope. It also presents the project charter, the tools and technologies used, and explains the work methodology and project timeline.

⌘ Chapter Two: Reference Study

This chapter reviews the most similar global applications and compares them with the public transport hub management system, highlighting what distinguishes these applications and the services offered by each system. Learning from these applications helps in developing the proposed system to provide added value.

⌘ Chapter Three: Analytical Study

This chapter addresses the functional requirements of the system and analyzes user interactions through process modeling using use case diagrams to ensure all required functions are covered. It also presents the characteristics of the actors and explains how the system can serve each category of actors.

⌘ Chapter Four: Design Study

This chapter explains how to build the overall platform architecture using the FLASK framework and a client–server layered architecture, focusing on task

distribution between the front-end and back-end. It establishes the logical system structure and technical foundations to ensure the system will not only meet the current phase requirements but will also be scalable and upgradable in future versions.

⌘ Chapter Five: Practical Implementation

This chapter presents the practical aspects of implementing the system. It documents how the predefined requirements are converted into real functionalities through programming, tested within an integrated development environment, and displays the system interfaces.

7- Basic Definitions and Concepts -

Chapter Two: Project Development and Management Methodology

Project Definition

- In city centers and service locations, especially for state officials, due to its great role in completing their tasks, as it depends on papers and old tools, it has been congested, and it is not a luxury but an urgent need.

Project objective

- It enhances the quality of life for citizens in the Damascus Governorate. The project includes designing and implementing a website to manage public transportation (buses) to and from the governorates. The main goal is to provide safe, reliable, and easily accessible transportation services to citizens, which contributes to reducing traffic congestion, improving the quality of transportation service, and enhancing

Scope

- Public Transportation

This system offers many important functions that help facilitate the management of public transportation and provide information. It allows the manager to automate transportation operations accurately, in terms of their status, quality, and requirements. It enables adding, deleting, or modifying trips, buses, or lines, managing tickets for each transportation line, and inquiring about schedule or route changes. These operations make public transportation centers useful.

This system includes many important functions, allowing citizens (passengers) to know the number and times of trips, use it easily, and don't forget the drivers, who can communicate with the garage manager to report an issue or file a complaint online, making the process more flexible. This system reduces traditional paper-based methods and processes.

1-Development Method Used (Iterative Incremental)

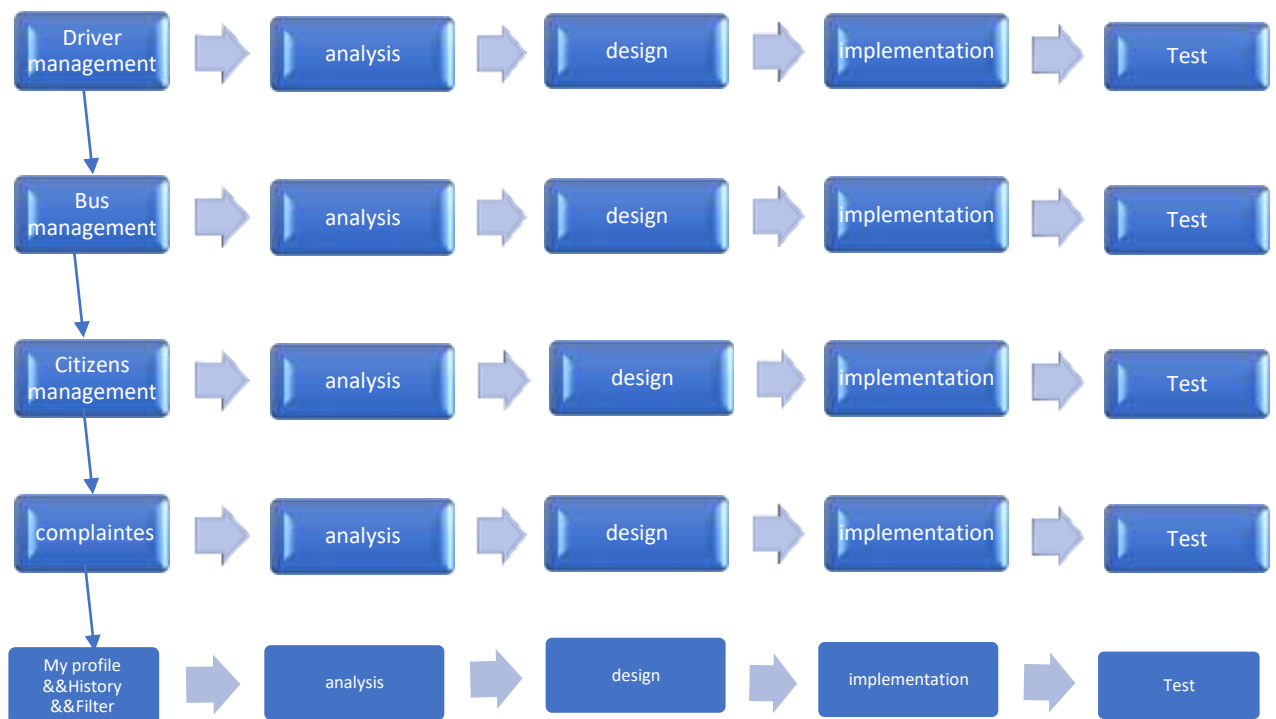
The incremental model is a development approach in which the system is divided into small functional parts (increments), with each part being developed and delivered sequentially until the system is complete.

1. Early and continuous delivery

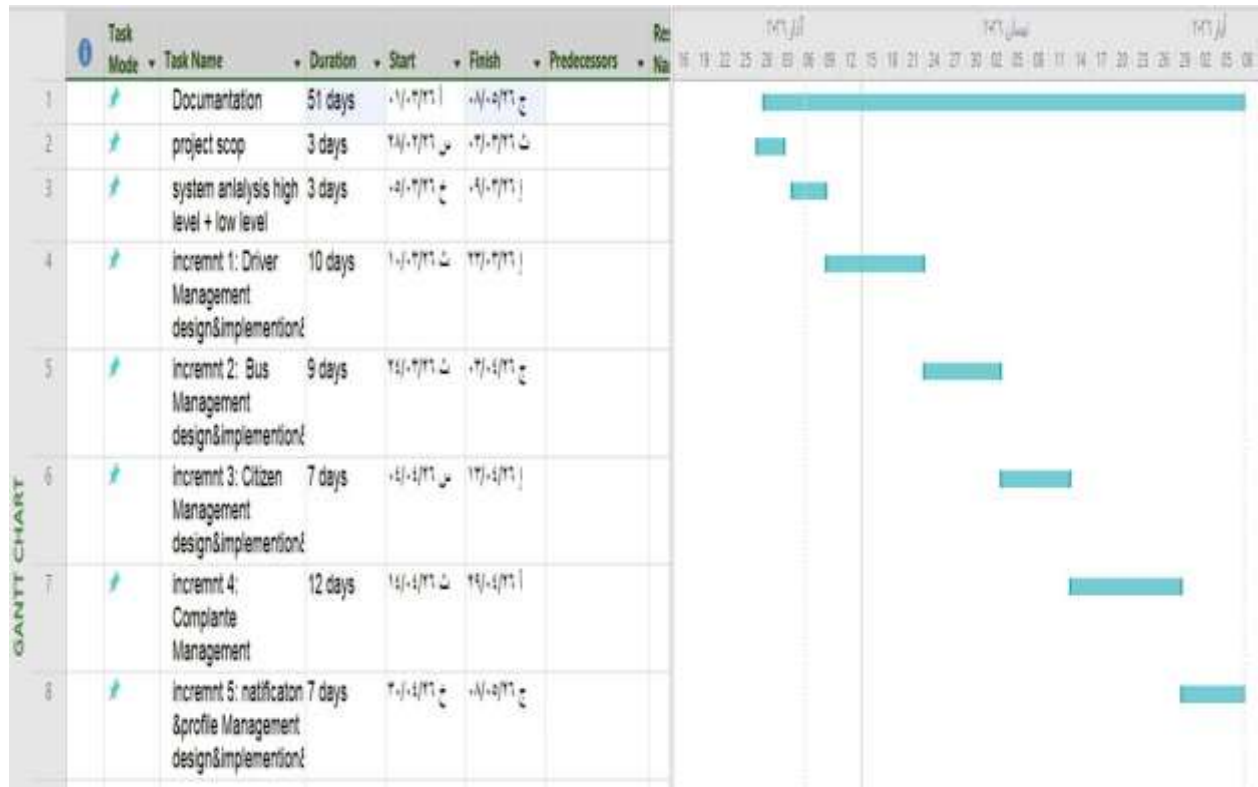
Flexibility and change management

Risk management

2. Parallel work



2-Work plan



Project charter

Project title	Public transport launch center management system
Project start date	28/2/2026
Project finish date	30/4/2026
Project manger	Automating and developing transportation and public transit processes in order to manage transport lines and work on organizing the operations conducted at the transport center (garage)
Project objectives	
Approach	1– Define the project scope and objectives. 2– Accurately analyze the system specifications and requirements. 3– Design and analyze the system architecture. 4– Develop the first phase with a focus on access permissions. 5– Test the functions of the first phase and verify its permissions. 6– Second phase: Integrate the system with the state's systems and institutions. 7– Conduct tests on the second phase. 8– Document all phases and their results.

7-Roles and Responsibilities

Name	Rule	Responsibilities
Eng.Rania Rajab Eng .Mohamad Al Balky	supervisor	Highly project management
Mohamad alnasser	Developer	Frontend Development
Ahmad Al Hafny	Developer	Back end
Ahmad Al kzhaly	Developer	Analyst and tester

8- Risk management

Risk Title	Risk Description	Raised Date	Tracking Frequency	State	Impact	Mitigation Plan
Delay in the schedule	Failure to adhere to the schedule due to development complexities or lack of resources.	05/3/2026	weekly	Active	High	Dividing the project into increments, using tools like a Gantt Chart, with prioritization of tasks .
Conflict in task distribution	Tasks are accumulating on one team member.	05/3/2026	weekly	Active	Medium	periodic meetings.
Difficulty Learning new techniques	The team is slow in mastering .NET Docker, which delays delivery.	10/3/2026	weekly	Active	High	Allocating weekly educational sessions.
Problems in interface integration	Conflict between Frontend (React) and Backend (.NET) during development.	10/3/2026	weekly	Active	High	Frequent integration testing, using detailed API documentation, and continuous coordination among team members.
Shortage Experts Testing	The appearance of undetected errors due to the absence of a specialized laboratory in the team.	01/04/2025	weekly	Active	High	Applying manual tests (Manual Testing) for each feature and assigning a member to understand the basics of testing.

Table 1 risk management

9- Project Quality Management

To ensure the delivery of a highly reliable software product, the following procedures were adopted:

- * Code Review: Periodically reviewing the code to ensure it follows best practices (Clean Code).

- * Preliminary Testing: Conducting unit tests for the software logic related to route calculations and bus definitions.

- * Acceptance Criteria: Defining precise criteria for each feature before considering it "completed" to ensure it is free of logical errors.

Project Management Tools Used

A set of digital tools was relied upon to organize workflow and document tasks:

- * GitHub Projects: to manage programming tasks and distribute them via a Kanban board.

- * Trello: to take quick notes and coordinate ideas related to application interfaces.

- * Google Sheets: to document spreadsheets, technical comparisons, and Gantt charts.

Chapter 3

Reference study

1- Similar applications

Three leading global platforms in the field of public transportation, passenger management, and two applications for private individual transport were selected.

- 1– Lyft
- 2– Carem
- 3– Moovit
- 4– Transit App
- 5– Citymapper

1.1–Lyft

- A transportation service app that allows users to request private cars to take them from one place to another. It was founded in 2012 in the United States, offering the following features:

1. Individual transportation service that provides private cars to transport individuals from their location to their destination quickly and easily.
2. Shared ride service that allows users to share the ride with other passengers, reducing the cost.
3. Provides larger cars that accommodate more passengers, ideal for families or groups.
4. Luxury car service that offers a premium transportation experience with professional drivers.

1.2–Careem

- Careem was founded in Dubai, United Arab Emirates, by Mudassir Sheikha, Abdullah Elyas, and Tariq Hashmi.

1. Smart App: It has an easy-to-use app that allows users to book and track rides easily.
2. Diverse Services: It offers a variety of services, including transportation services.
3. Transportation Service: It provides private cars for transporting individuals, in addition to options such as Careem GO and Careem XL for groups.
4. Fast Delivery: It offers the Careem NOW service, which allows food to be delivered quickly from restaurants to homes.

1.3–Moovit

1. A global app that provides trip planning.
2. Supports real-time bus tracking.
3. Provides text and audio data for the stations.
4. Relies on artificial intelligence algorithms to reduce waiting time.

1.4–Transit App

1. An app that provides maps and transportation routes
2. Supports real-time notifications for line changes
3. Offers estimated arrival times
4. Used by millions of users in Europe and America

1.5–CityMapper

1. Supports trip planning by integrating more than one mode of transport (bus – microbus – metro)
2. Interactive user interface
3. Comparison of time and cost between transportation options
4. Provides usage reports

2-Comparison table between systems

SYSTEM OVERVIEW

Feature Comparison — Our System vs. Market Leaders

Feature	Moovit	Transit	Citymapper	Our System ★
General Trip Info	✓	✓	✓	✓
Real-time Tracking	✓	✓	✓	X
Multi-modal Planning	✓	✓	✓	—
Reservations & Tickets	✓	✓	✓	—
Garage Operations Mgmt	—	—	—	✓★
Driver & Bus Management	—	—	—	✓★
Delay Notifications	✓	✓	✓	✓
Admin Interface	—	—	—	✓★
Data Analytics	✓	✓	✓	X
Map Integration	✓	✓	✓	X

Chapter 4

Analytical study

1-Feasibility study

2-Requirements Elicitation

We carried out the process of deriving the project requirements through the theoretical study of the project, as well as through repeated meetings with the stakeholders (Daraa Garage Manager) and the assistance of our supervisor. The requirements were also derived and collected through the help of studying similar projects.

3-stakeholders

- 1– **Bus Drivers** :responsible for updating trip ,status , completing routes, and repor ting issues during transit .
- 2– **Garage manager** :manages drivers , buses , routes ,trips ,and handles Citizen complaints and and operations .
- 3– **Passengers** : citizens who book tickets , track trips , view schedules, and submit complaints .

4-functional requirements

FB1:user can be log in

FB2:user can be mange account

FB3: user can be search with filter

FB4: Garage manger can be add new vehicle

FB5:Garage manger can be update vehicle

FB6:Garage manger can be delete vehicle

FB7: Garage manger can be search vehicle

FB8: Garage manger can be add new citizen

FB9; Garage manger can be update information citizen

FB10: Garage manger can be delete citizen

FB12 : Garage manger can be search citizen

FB13 Garage manger can be add new driver

FB14: Garage manger can be update information driver

FB15 : Garage manger can be delete driver

FB16 : Garage manger can be search driver

FB17 : driver and citizen can be send complaints

FB18: all actors can be show complaints

FB19: Garage manger can be respond complaints

FB 20 ;All actors can be view history

FB21 : all actors can be search with filter ;

5-Nonfunctional requirement

Nonfunctional requirement	
Security	Authentication using username and password with encryption of user data using a hash algorithm Implementing permission granting in the authorization system
Performance	The system response time must not exceed 2 seconds when executing basic operations.
Usability	The system interfaces must be simple. The user should be able to complete basic operations (such as booking a trip) easily without training. The system should be usable across different devices (computer, tablet, mobile).
Availability	The system must maintain operational readiness to ensure continuous availability of system services.

6–Prioritization of requirements and the mechanism for their management

ID	Requirement Name	Description	Actor(s)	MoSCoW	Justification
1	Driver Management	Manage drivers (add, edit, delete, search)	Garage Manager	Must	Core system functionality
2	Bus Management	Manage buses (add, edit, delete, search)	Garage Manager	Must	Essential for operations
3	Citizens Management	Manage citizens data	Garage Manager	Must	Required for service delivery
4	Complaints Management	Handle complaints (send, respond, view)	All Actors	Must	Critical for user interaction
5	Profile Management	Edit and view user profile	All Actors	Should	Improves user experience
6	History	View system activity history	All Actors	Should	Useful but not critical
7	Search & Filter	Search and filter data	All Actors	Could	Enhancement feature
8	Mobile Application	Mobile version of the system	–	Won't	Outside the scope of the current project
9	GPS Integration	Real-time bus tracking	–	Won't	Needs additional resources

7-system description of features(High level use case diagram)

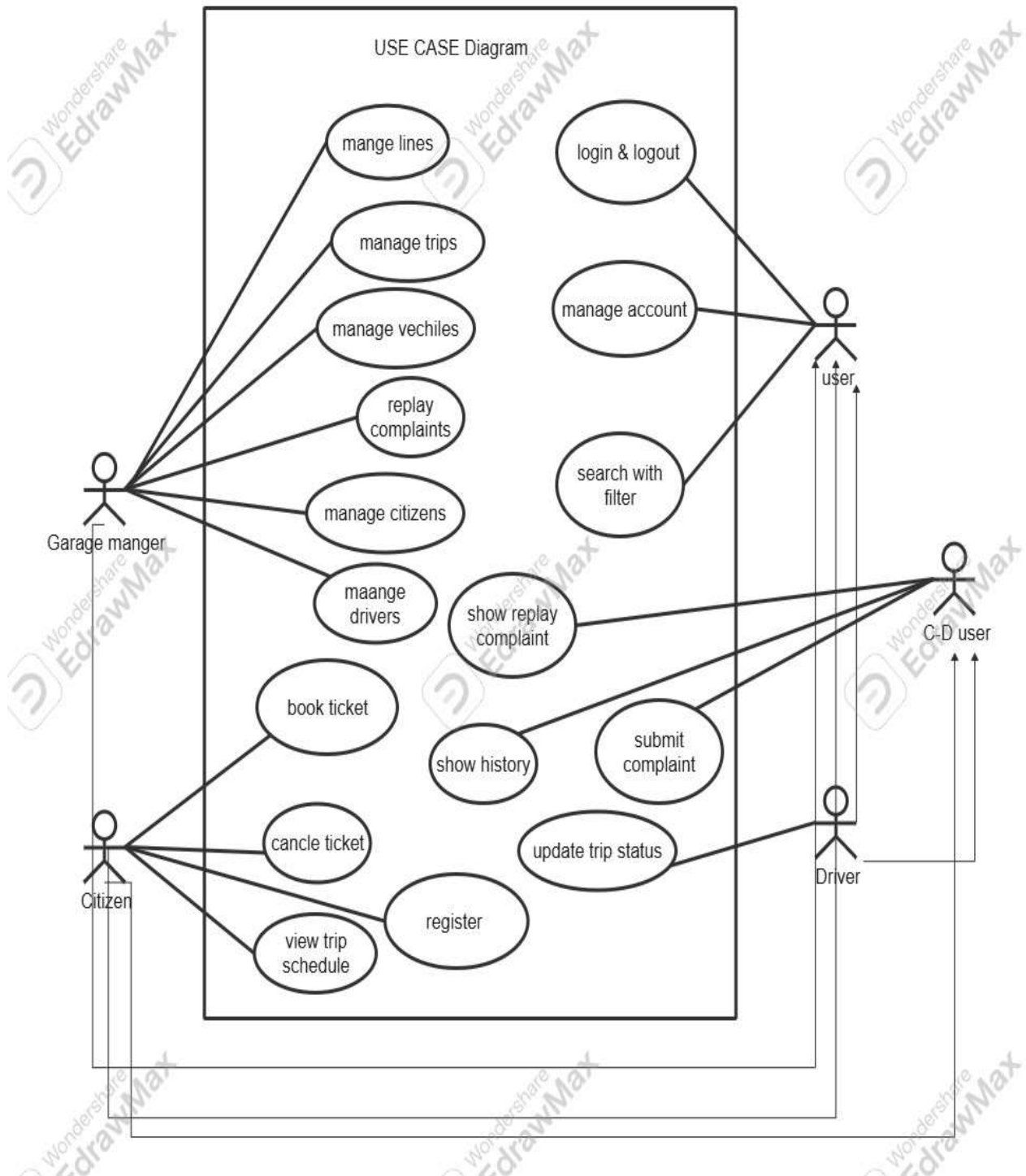


Figure 1 use case high level

7-1-Main Use Case Diagrams

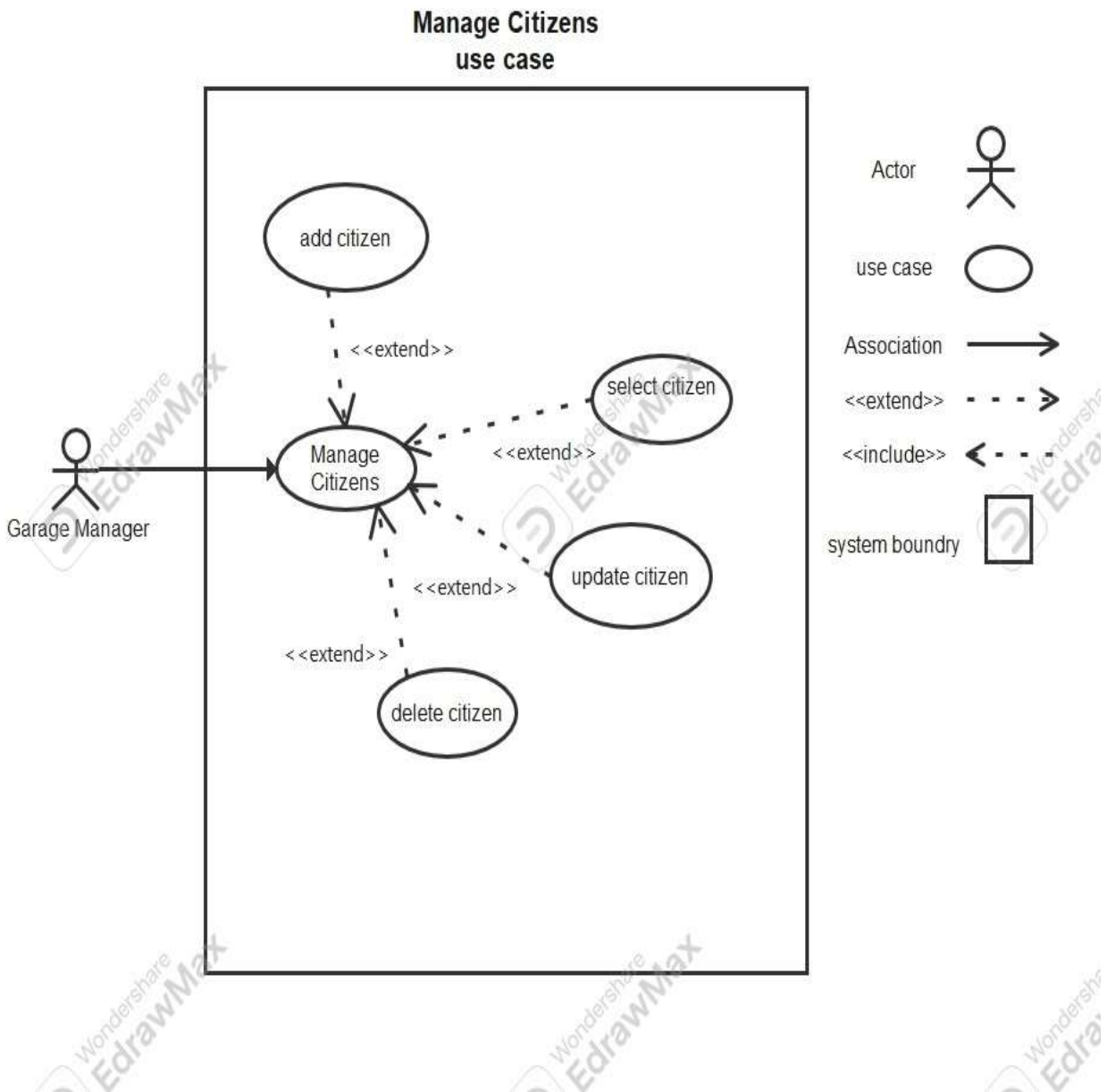


Figure 2 use case mange citizens

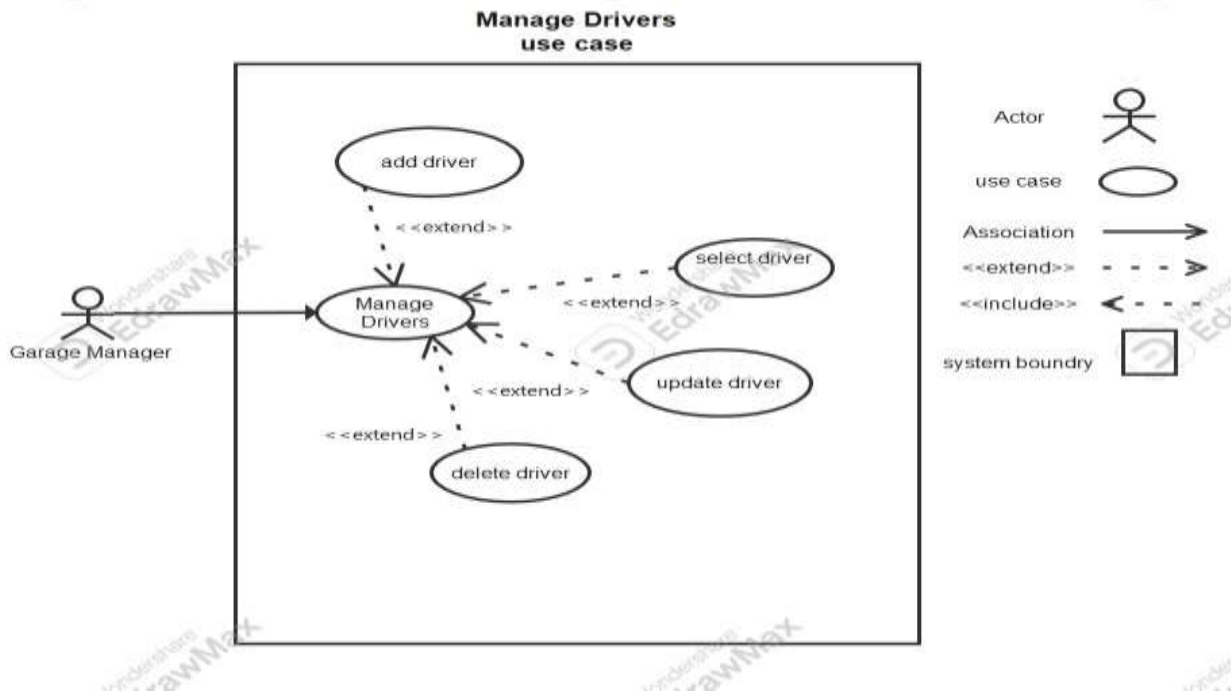


Figure 3 use case mange drivers

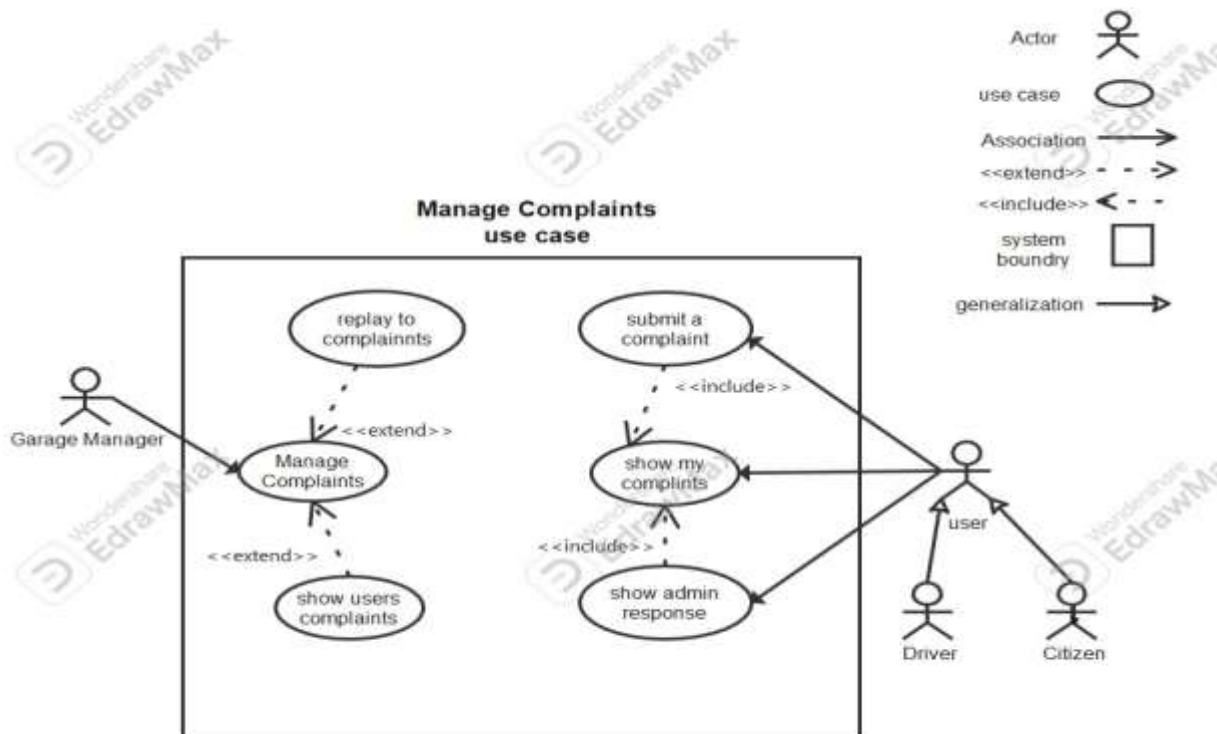


Figure 4 use case mange clmplaint

7-2- use case Descriptions

Field	Description
Use Case ID	UC-01
Name	Add Driver
Actor(s)	Garage Manager
Preconditions	– User is logged into the system – Dashboard is loaded– User is on "Manage Drivers" page
Postconditions	– New driver is created and stored in the system – OR Error message is displayed
Main Flow	1. User clicks "Manage Drivers" 2. System displays drivers page 3. User clicks "New Driver" 4. System displays driver form 5. User enters required information: – Username, Email, Phone Number – License Number, Issuing Authority – Vehicle Plate, Model, Color – License Expiry, Registration Expiry – License Category – Initial Password 6. User clicks "Create Driver" 7. System validates data 8. System saves driver 9. System displays success message: "Driver created successfully"
Alternative Flow	A1: Cancel Operation – User clicks "Cancel"– System returns to Manage Drivers page A2: Validation Error – Invalid or duplicate data (e.g., email exists) – System displays error message (e.g., "Email already in use")

Figure 5 description UC-01

Field	Description
Use Case ID	UC-02
Name	Update Driver
Actor(s)	Garage Manager
Preconditions	– User is logged in – User is on Manage Drivers page – Driver already exists
Postconditions	– Driver information is updated OR – No changes are made
Main Flow	1. User clicks "Edit" on a driver 2. System displays edit form with existing data 3. User modifies driver information: – Username, Phone Number – License details – Vehicle details – Expiry dates – License category 4. User clicks "Update Driver" 5. System asks for confirmation 6. User confirms update 7. System saves changes 8. System displays: "Driver updated successfully"
Alternative Flow	A1: Cancel Update – User clicks "Cancel"– System discards changes A2: Validation Error– Invalid data entered – System shows error message A3: User Rejects Confirmation – User cancels confirmation dialog– No update occurs

Figure 6 description UC-02

Table 2

Field	Description
Use Case ID	UC-03
Name	Delete Driver
Actor(s)	Garage Manager
Preconditions	– User is logged in – User is on Manage Drivers page – Driver exists
Postconditions	– Driver is deleted from the system OR – Driver remains unchanged
Main Flow	1. User clicks "Delete" on a driver 2. System displays confirmation message: "Are you sure you want to delete this driver?" 3. User clicks "Delete" 4. System removes driver from database 5. System displays: "Driver deleted successfully"
Alternative Flow	A1: Cancel Delete – User clicks "Cancel" – System keeps driver unchanged A2: Error During Deletion – System fails to delete driver – System displays error message

Figure 7 description UC-03

Field	Description
Use Case ID	UC-04
Name	Show Driver Details
Actor(s)	Garage Manager
Preconditions	- User is logged in - User is on Manage Drivers page - Driver exists
Postconditions	- Driver details are displayed
Main Flow	1. User clicks "Show Details" on a driver 2. System navigates to driver details page 3. System displays full driver information
Alternative Flow	A1: Driver Not Found - Driver does not exist - System displays error message

Figure 8 description

Field	Description
Use Case ID	UC-05
Name	Add Vehicle
Actor(s)	Garage Manager
Preconditions	– User is logged into the system – Dashboard is loaded – User is on "Manage Vehicles" page
Postconditions	– New vehicle is added to the system OR – Error message is displayed
Main Flow	1. User navigates to "Manage Vehicles" 2. System displays vehicles page 3. User clicks "New Vehicle" 4. System displays vehicle form 5. User enters vehicle information: – Name – Type – Capacity – Plate Number – Account Status (Active / Inactive) 6. User clicks "Create Vehicle" 7. System validates input 8. System saves vehicle 9. System displays success message: "Vehicle created successfully"
Alternative Flow	A1: Cancel Operation – User clicks "Cancel" – System returns to Manage Vehicles page A2: Validation Error – Invalid or duplicate data (e.g., plate number exists) – System displays error message

Figure 9 description UC-05

Field	Description
Use Case ID	UC-06
Name	Update Vehicle
Actor(s)	Garage Manager
Preconditions	– User is logged in – User is on Manage Vehicles page – Vehicle exists
Postconditions	– Vehicle information is updated OR – No changes are made
Main Flow	1. User clicks "Edit" on a vehicle 2. System displays edit form with existing data 3. User updates vehicle information: – Name – Type – Capacity – Plate Number – Account Status 4. User clicks "Update Vehicle" 5. System displays confirmation message: "Confirm Update: Are you sure you want to update this vehicle?" 6. User clicks "Update" 7. System saves changes 8. System displays: "Vehicle updated successfully"
Alternative Flow	A1: Cancel Delete – User clicks "Cancel" – System keeps vehicle unchanged A2: Error During Deletion – System fails to delete vehicle – System displays error message

7-3-Activity Diagrams

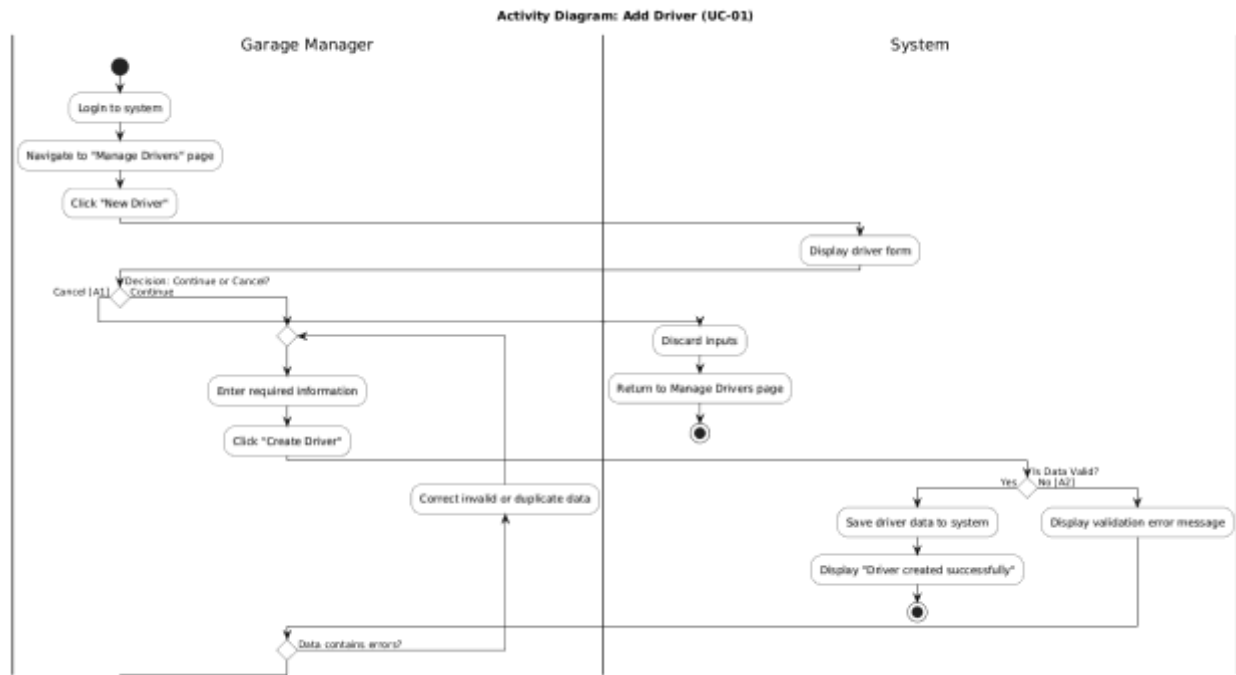


Figure 10 activity UC-1



Figure 11 activity UC 2

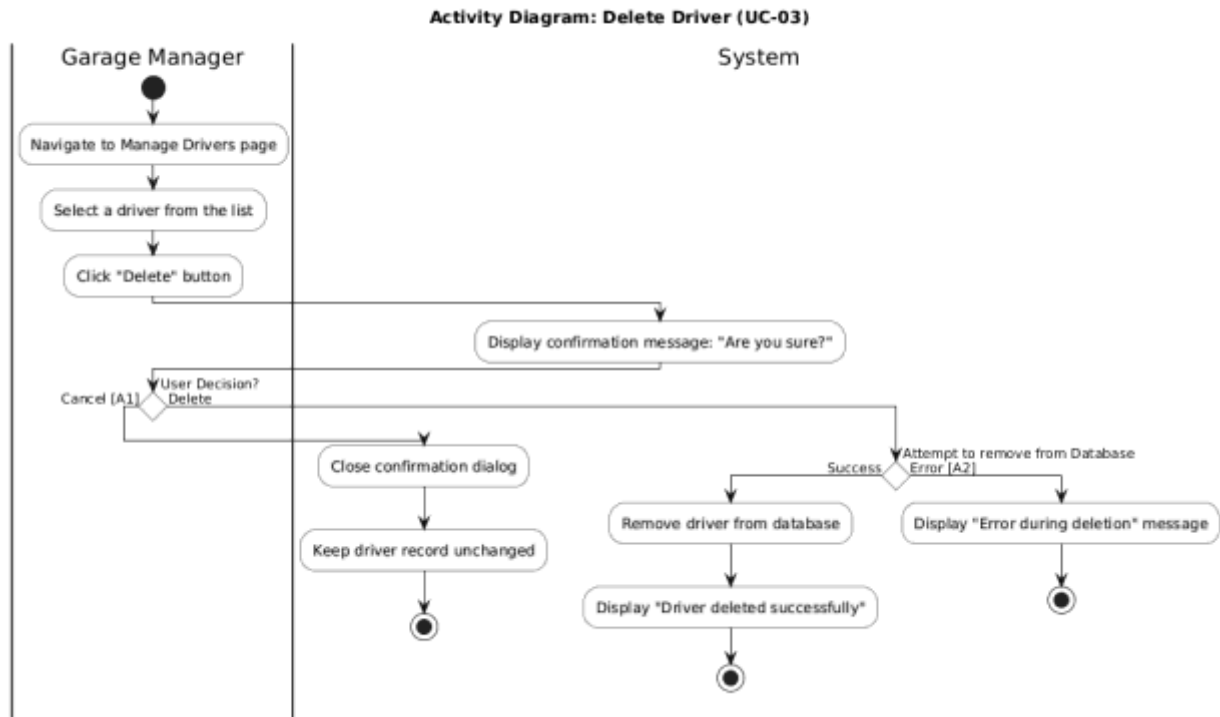


Figure 12activity UC-03

Activity Diagram: Show Driver Details (UC-04)

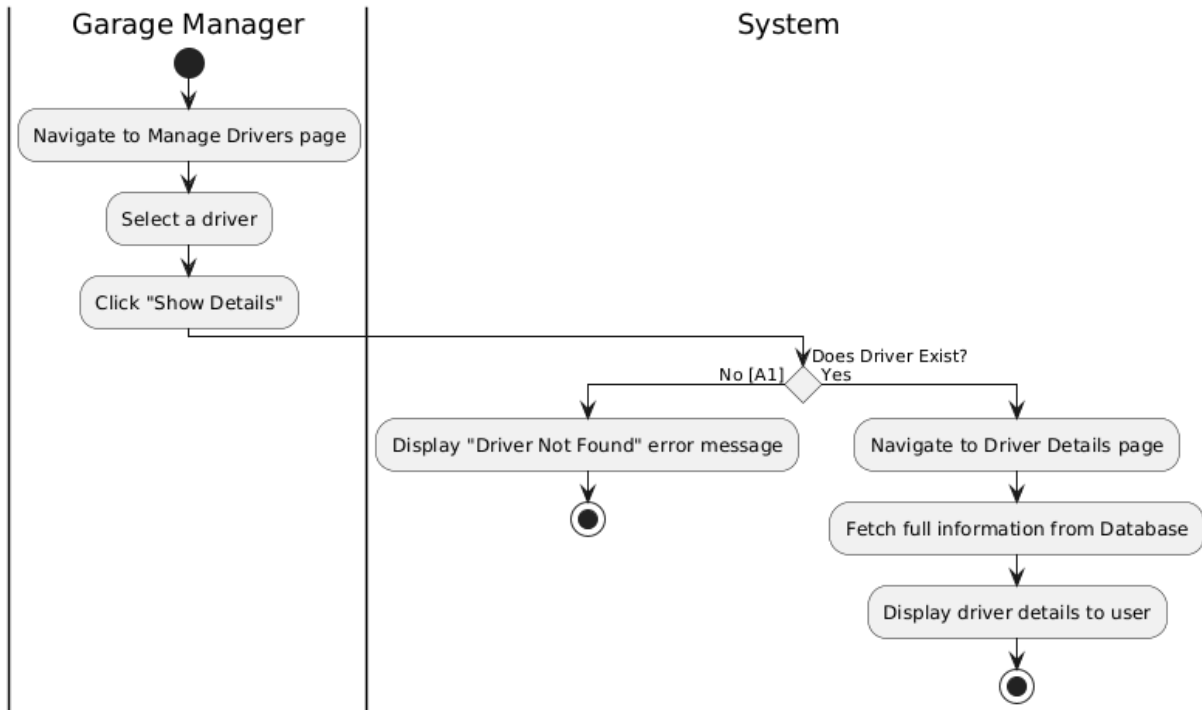


Figure 13 activity UC-4

Activity Diagram: Add Vehicle (UC-05)

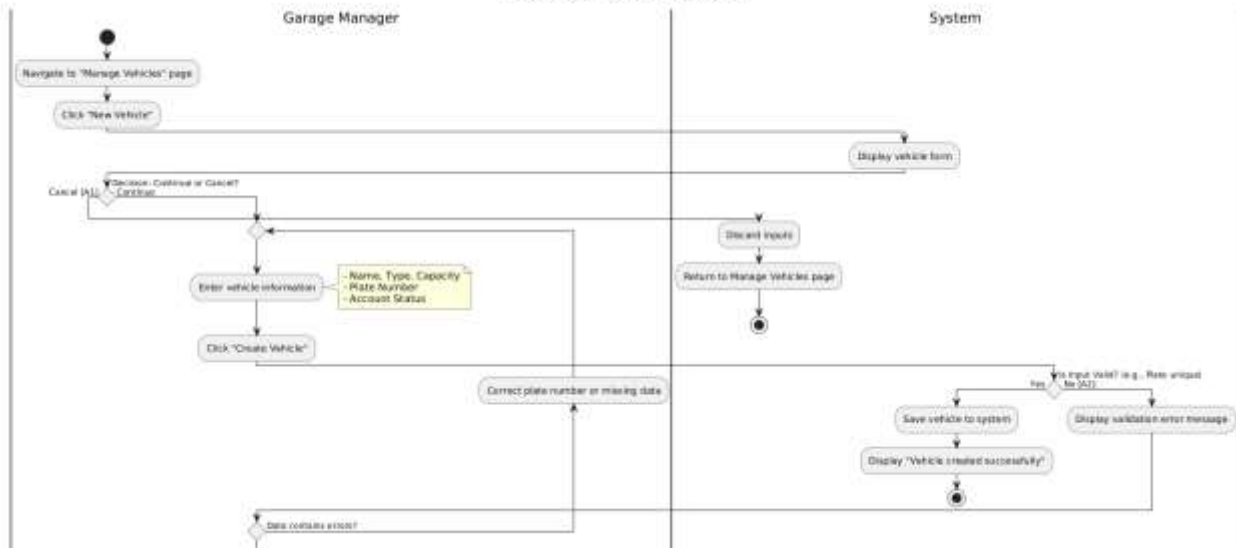


Figure 14 activity UC-05

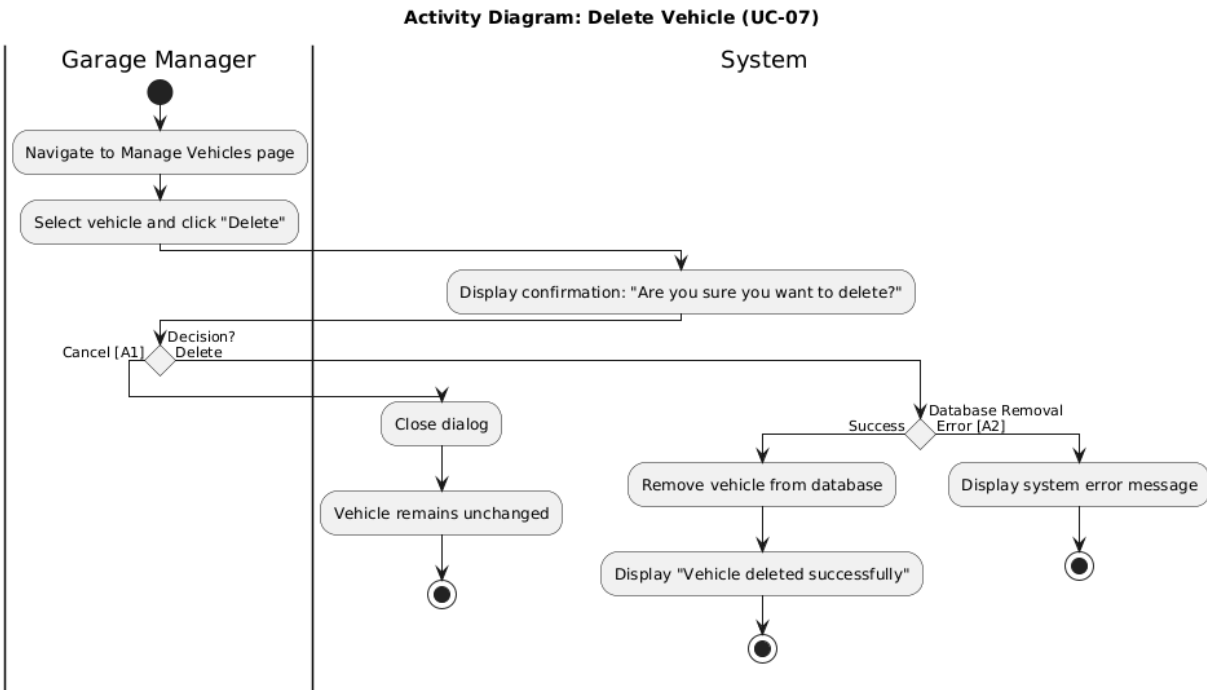


Figure 16 activity UC=7

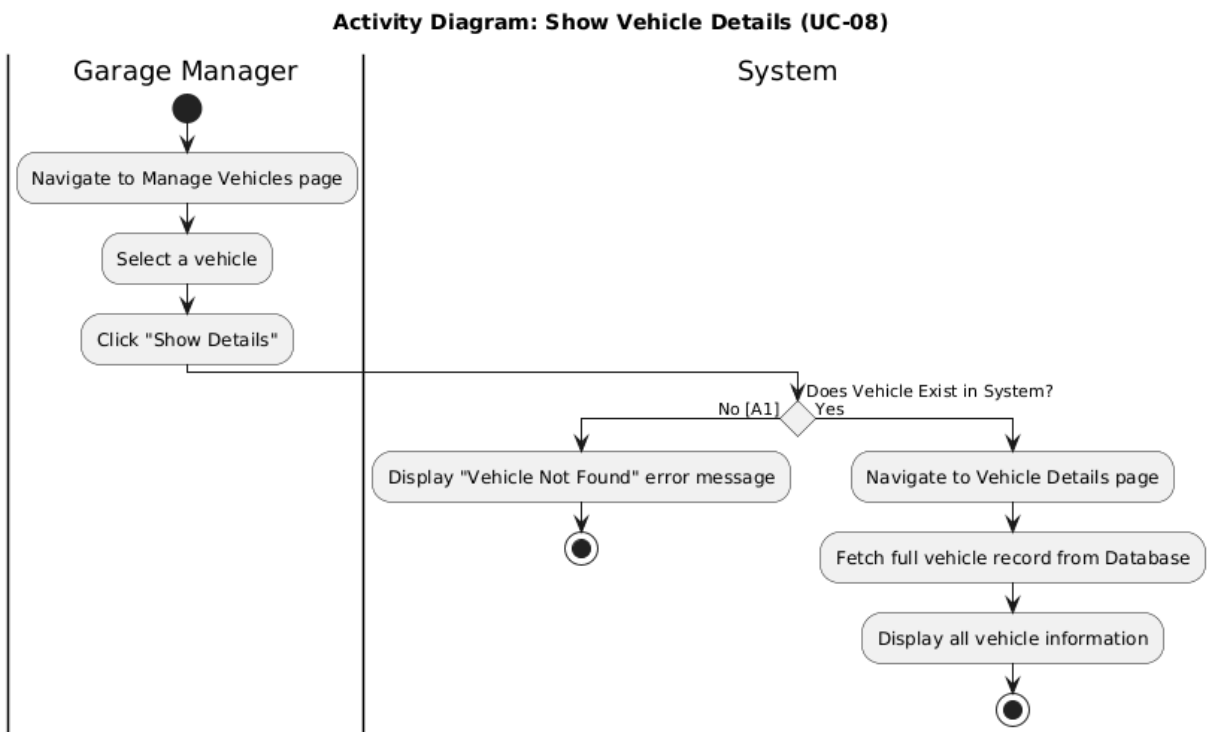


Figure 17 activity UC-08

Figure 18 activity UC-09

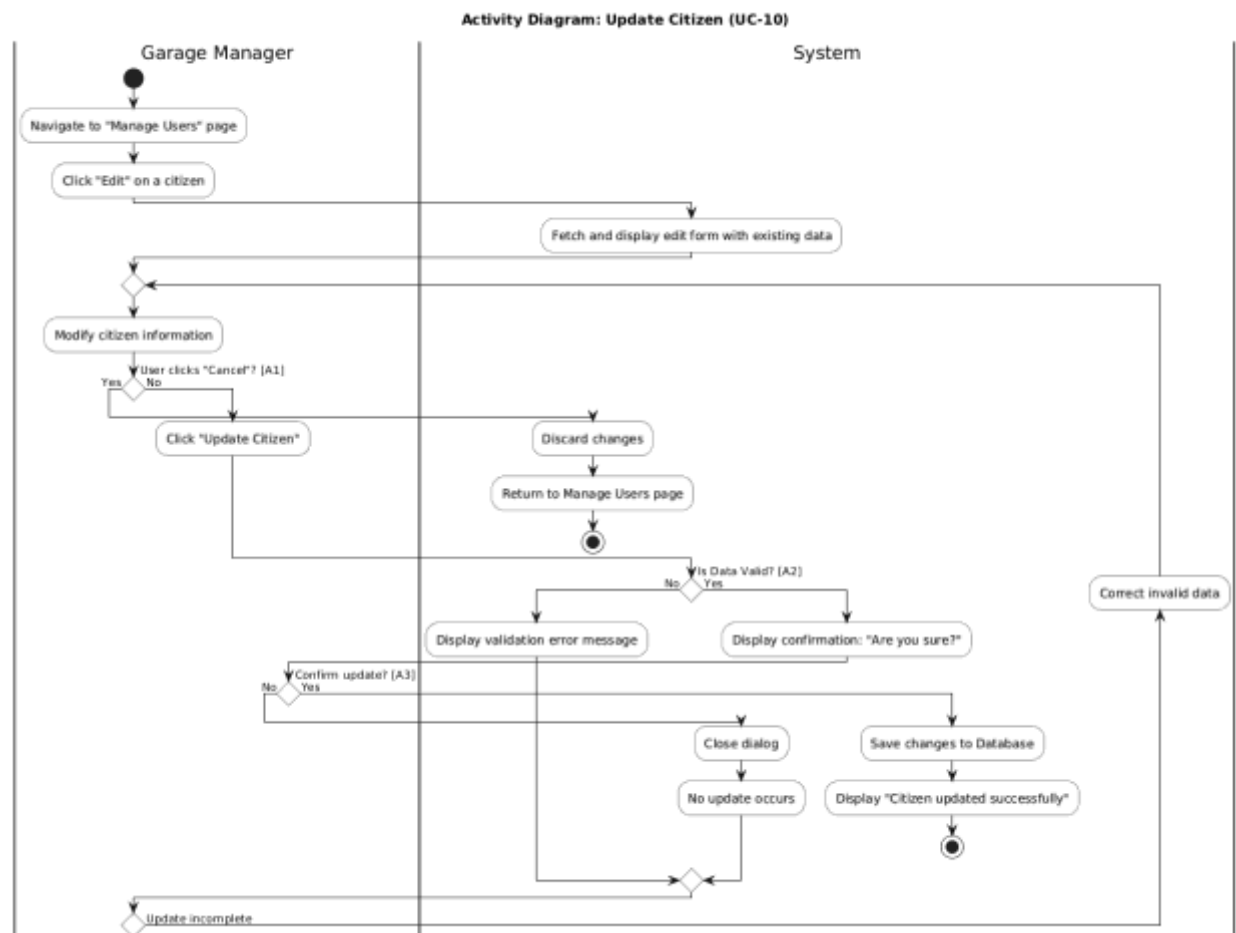
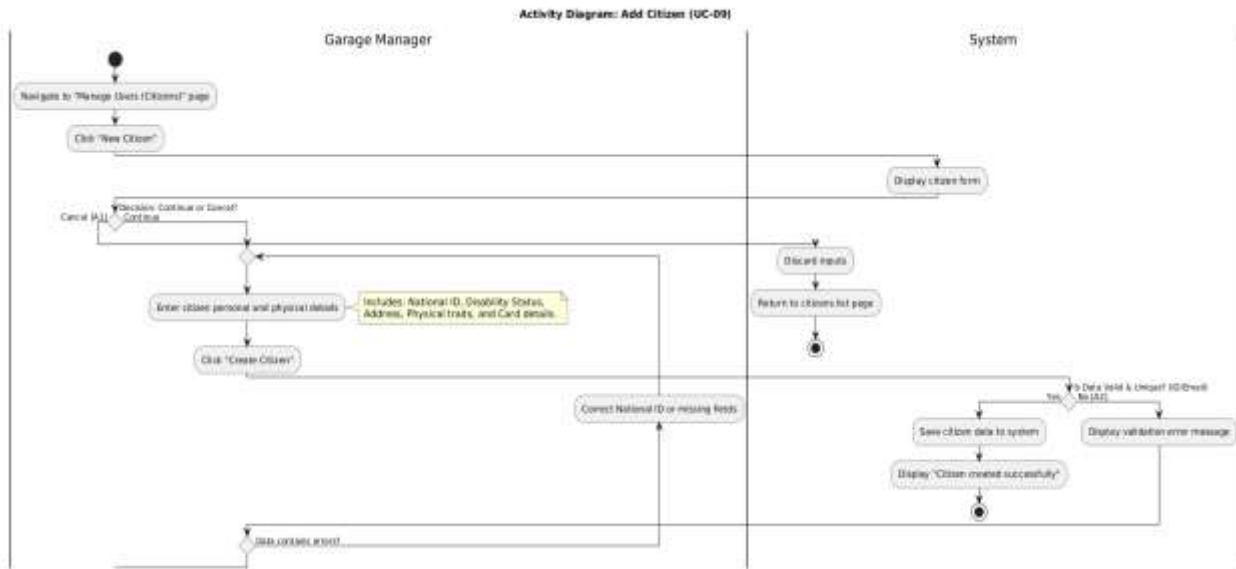


Figure 19 activity UC-10

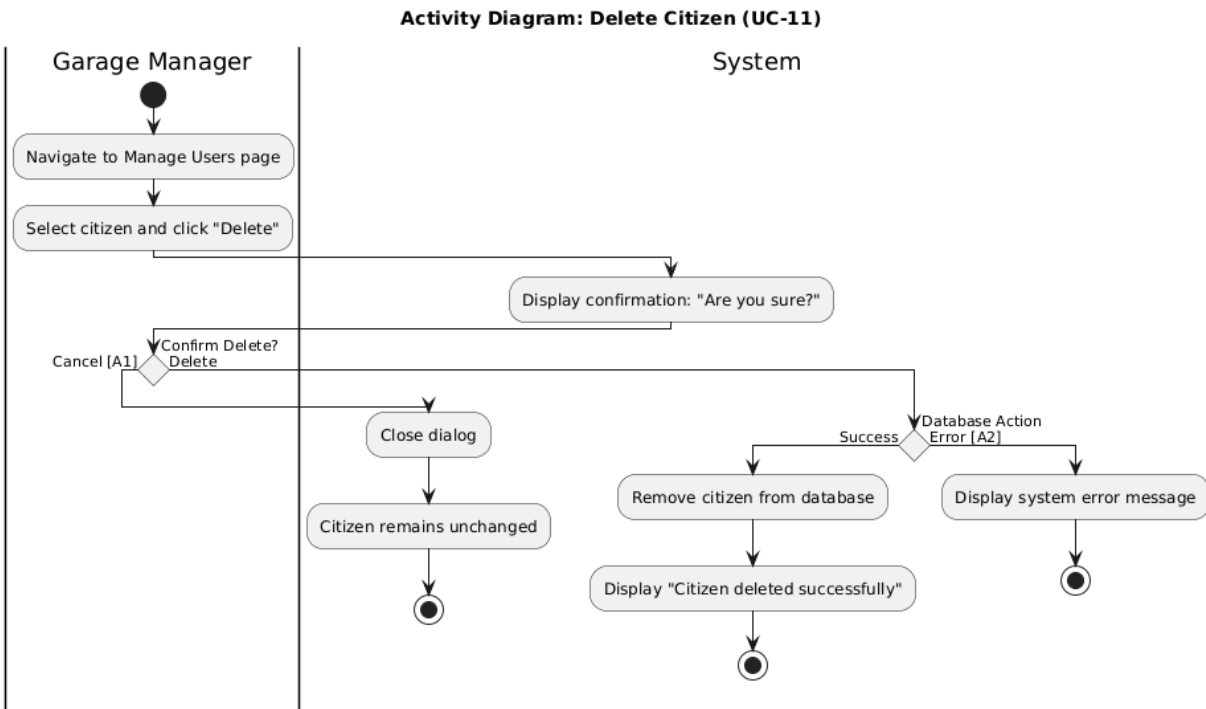


Figure 20 activity UC-11

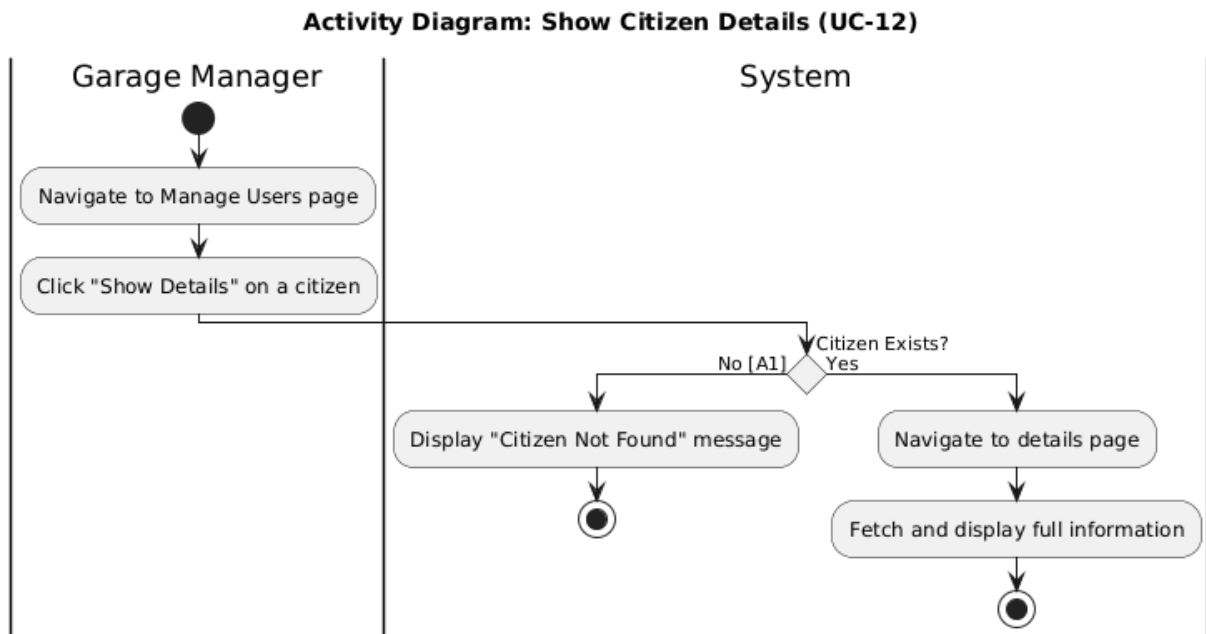


Figure 21 activity UC-12

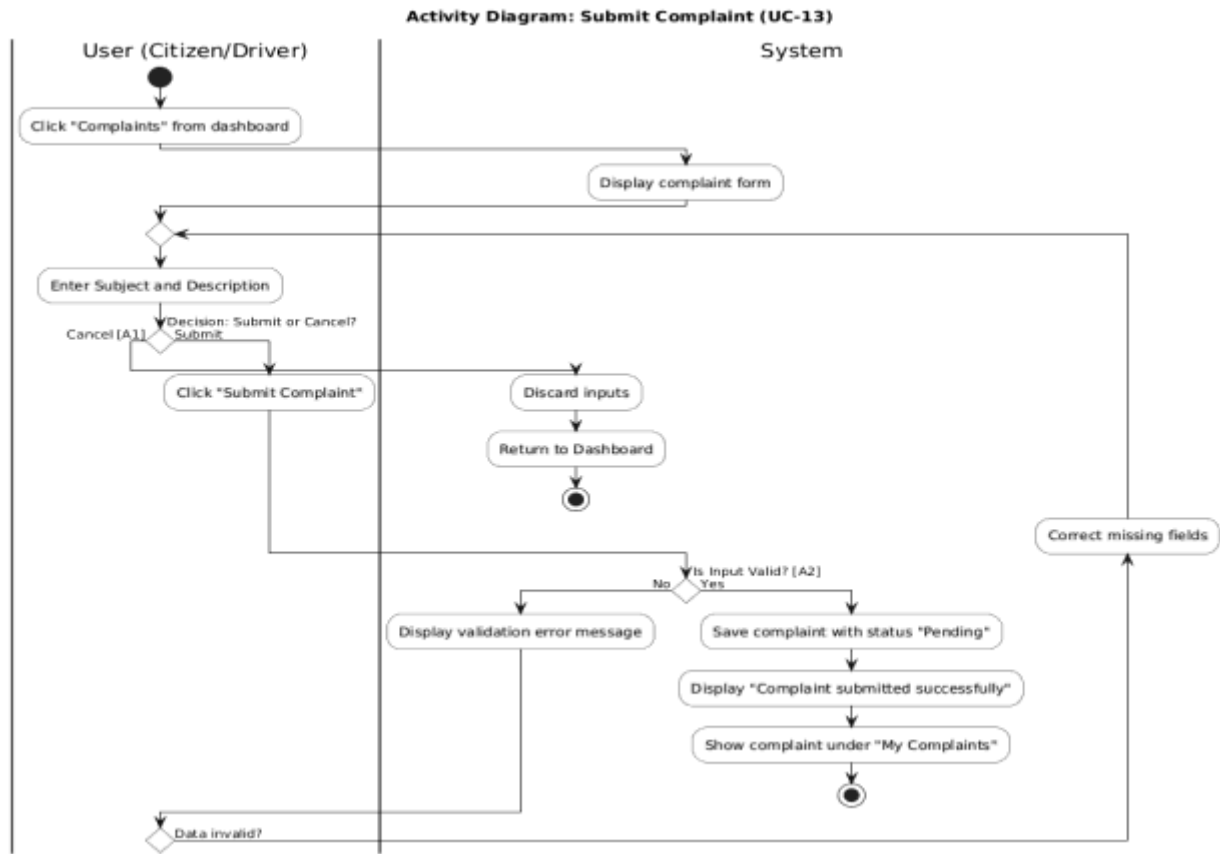


Figure 22 activity UC-13

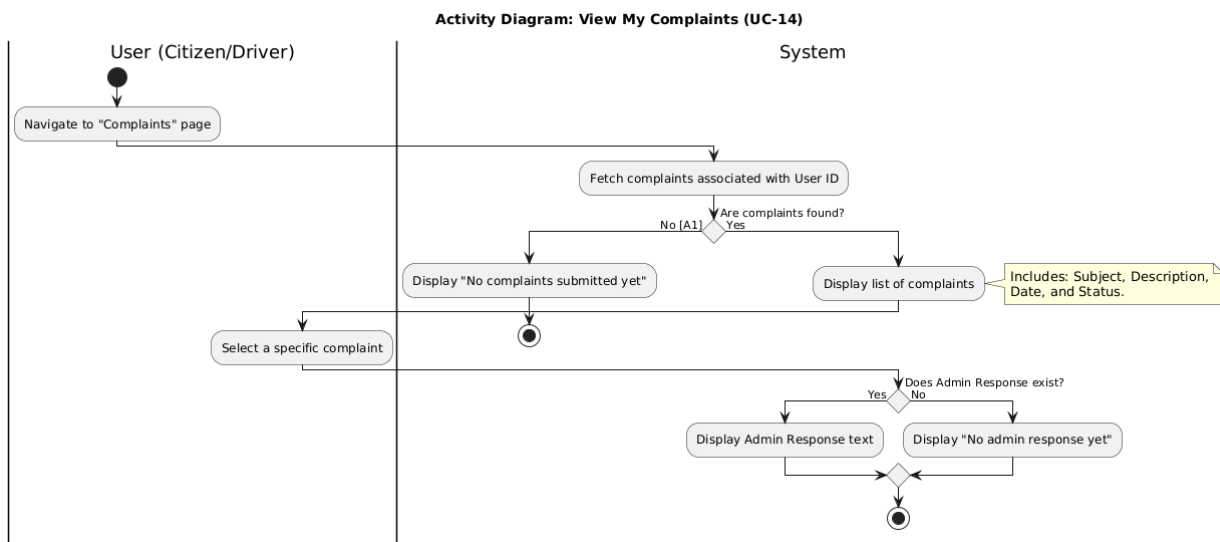


Figure 23 activity UC-14

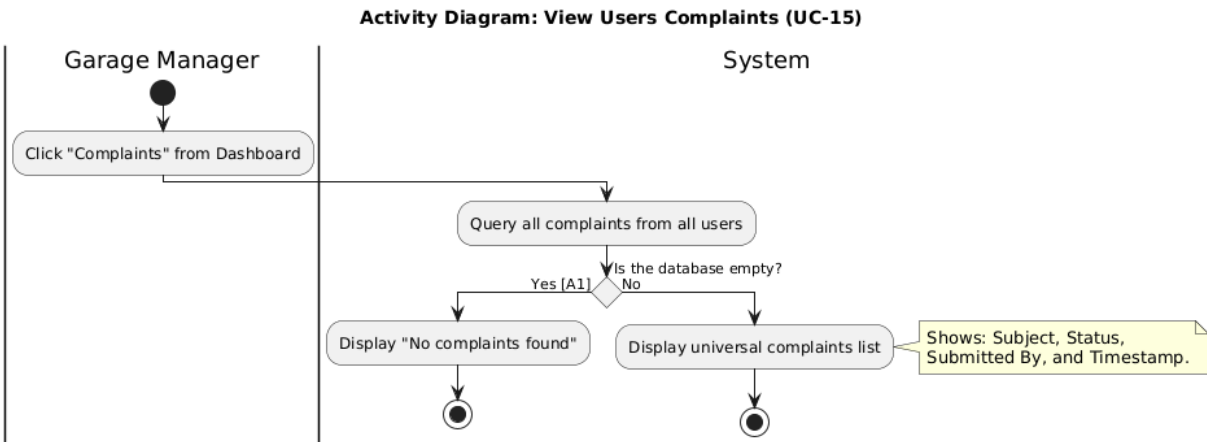


Figure 24 activity UC-15

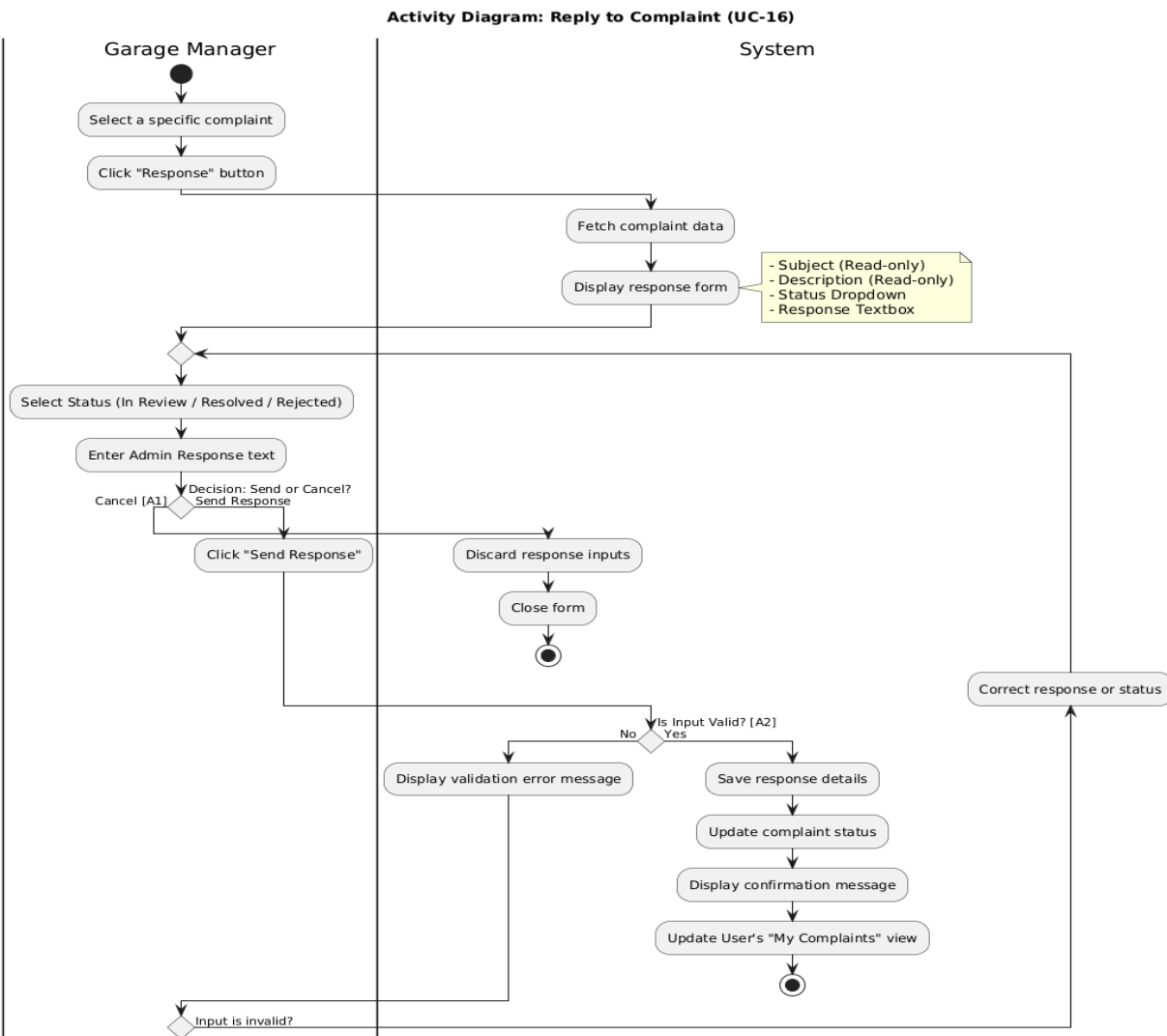


Figure 25 activity UC-16

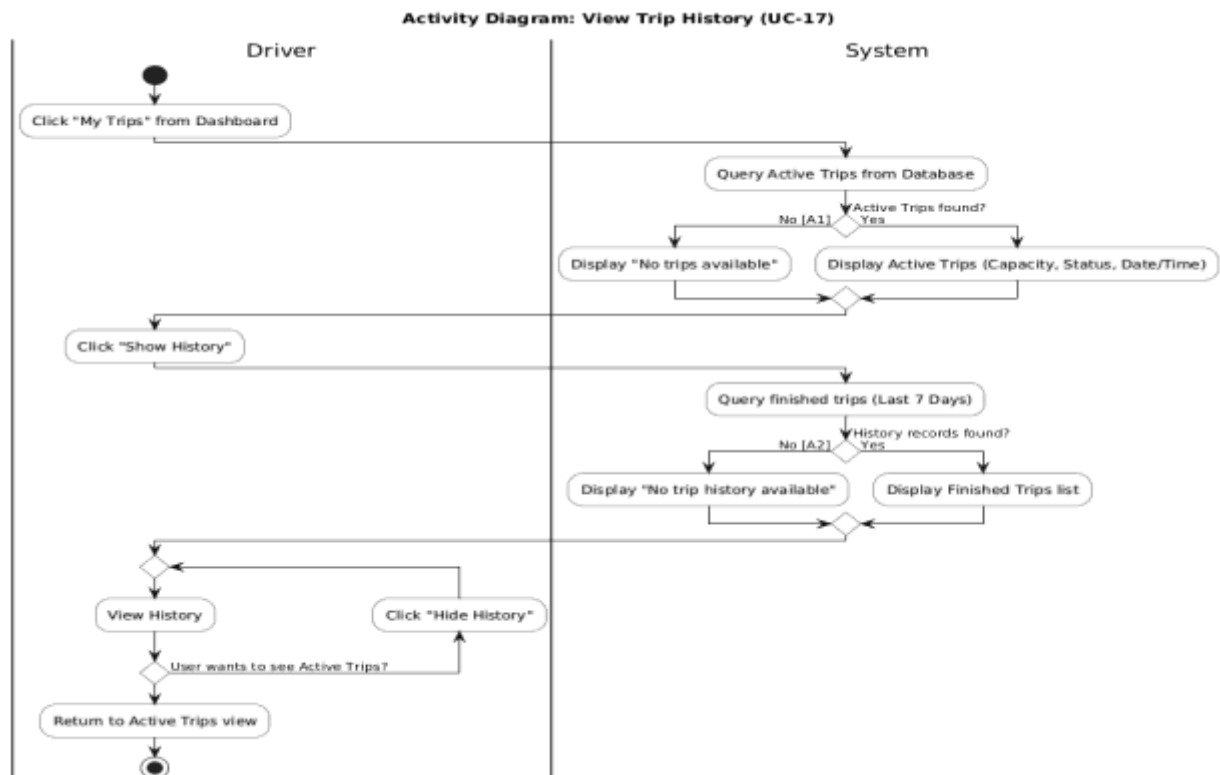


Figure 26 activity UC-17

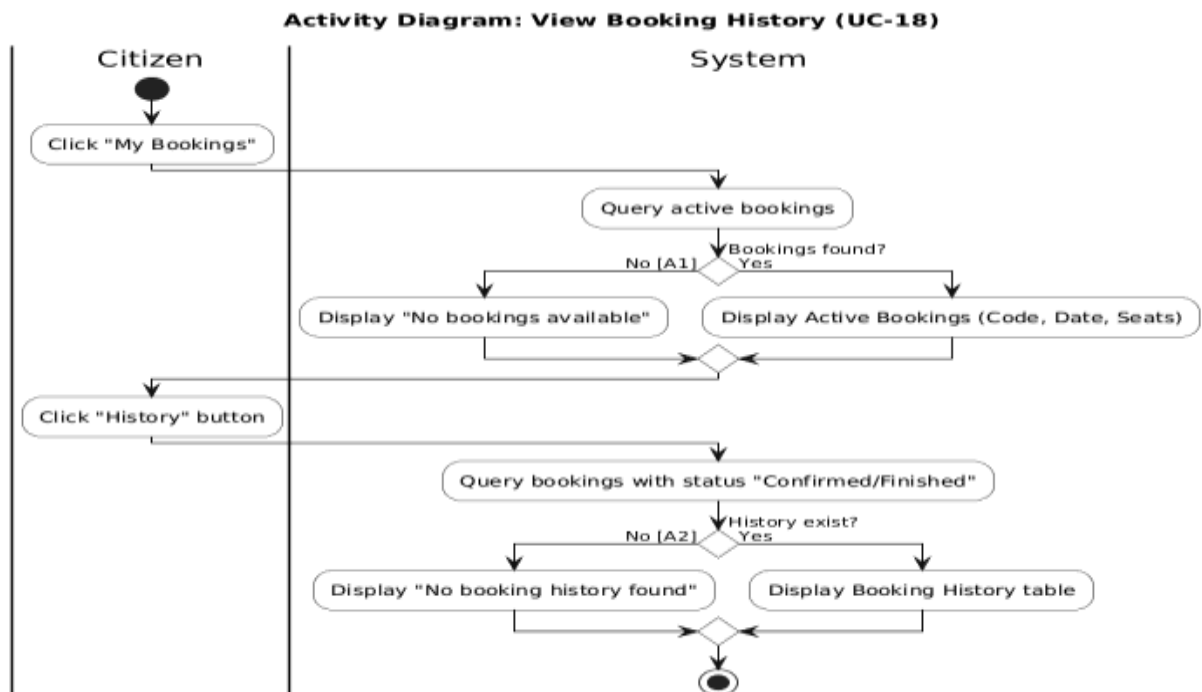


Figure 27 activity UC-18

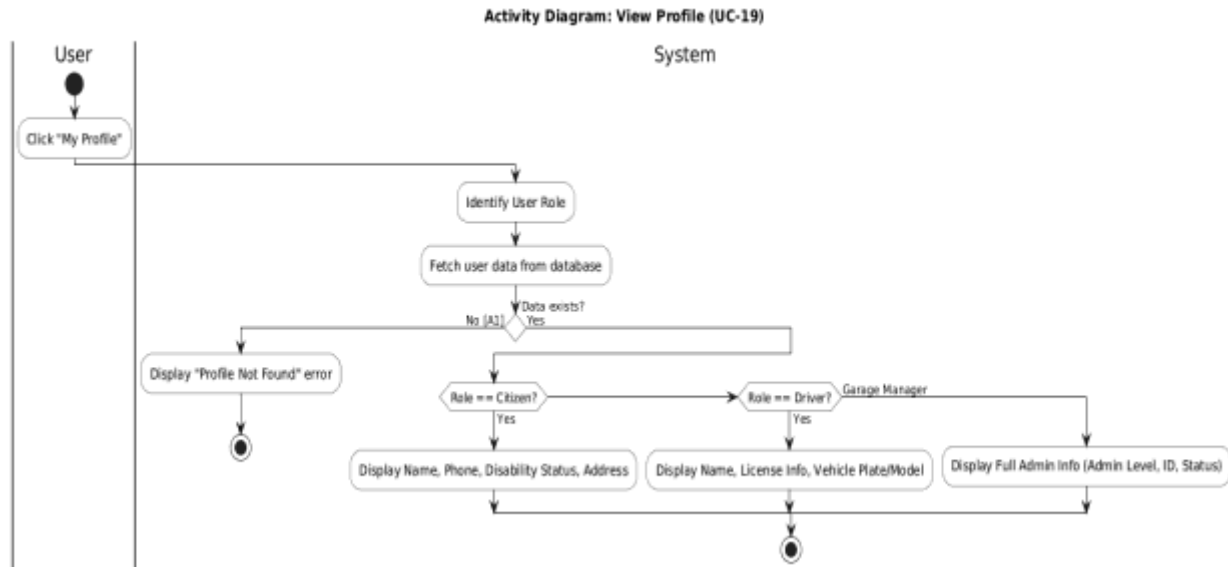


Figure 28 activity UC -19

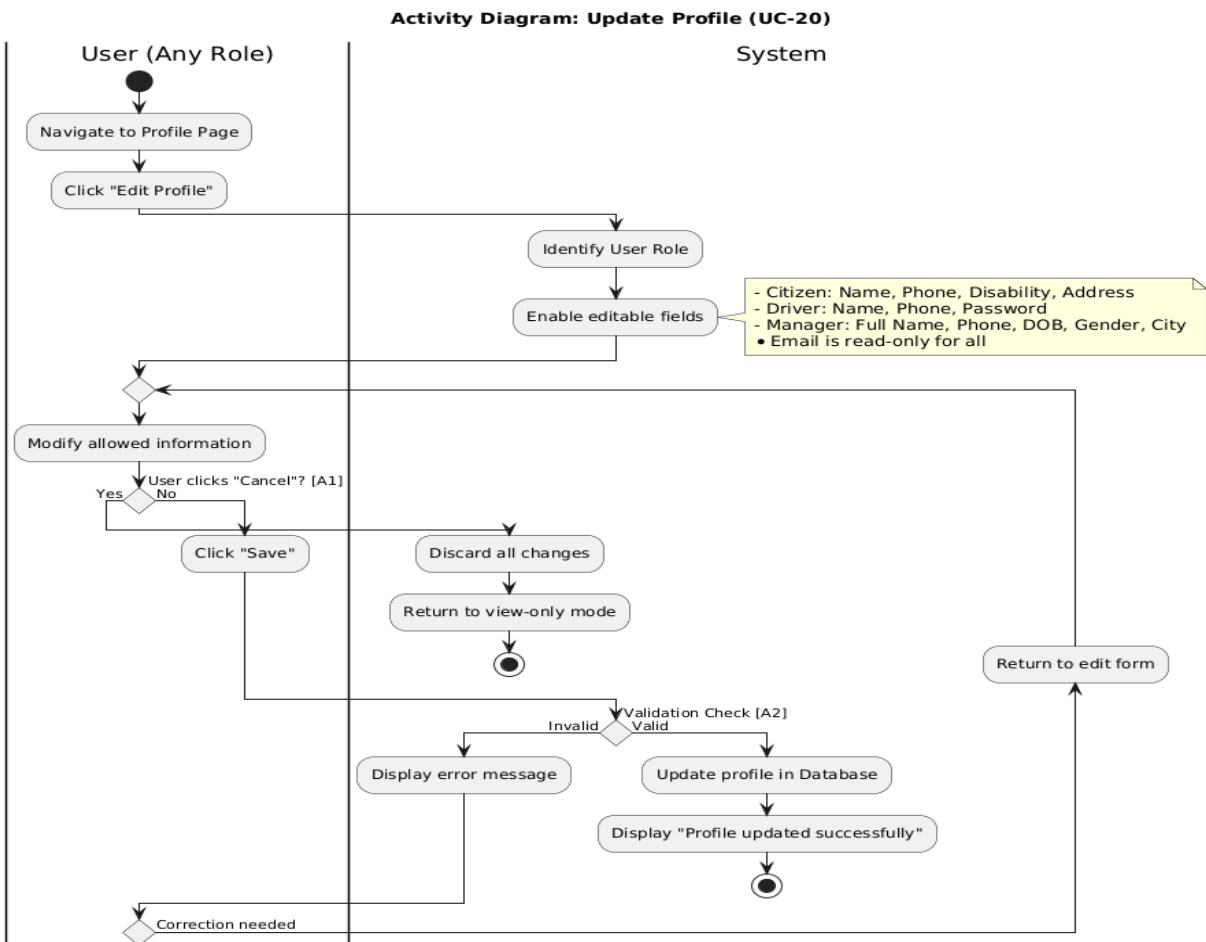


Figure 29 activity UC-20

Activity Diagram: Search with Filters (UC-21)

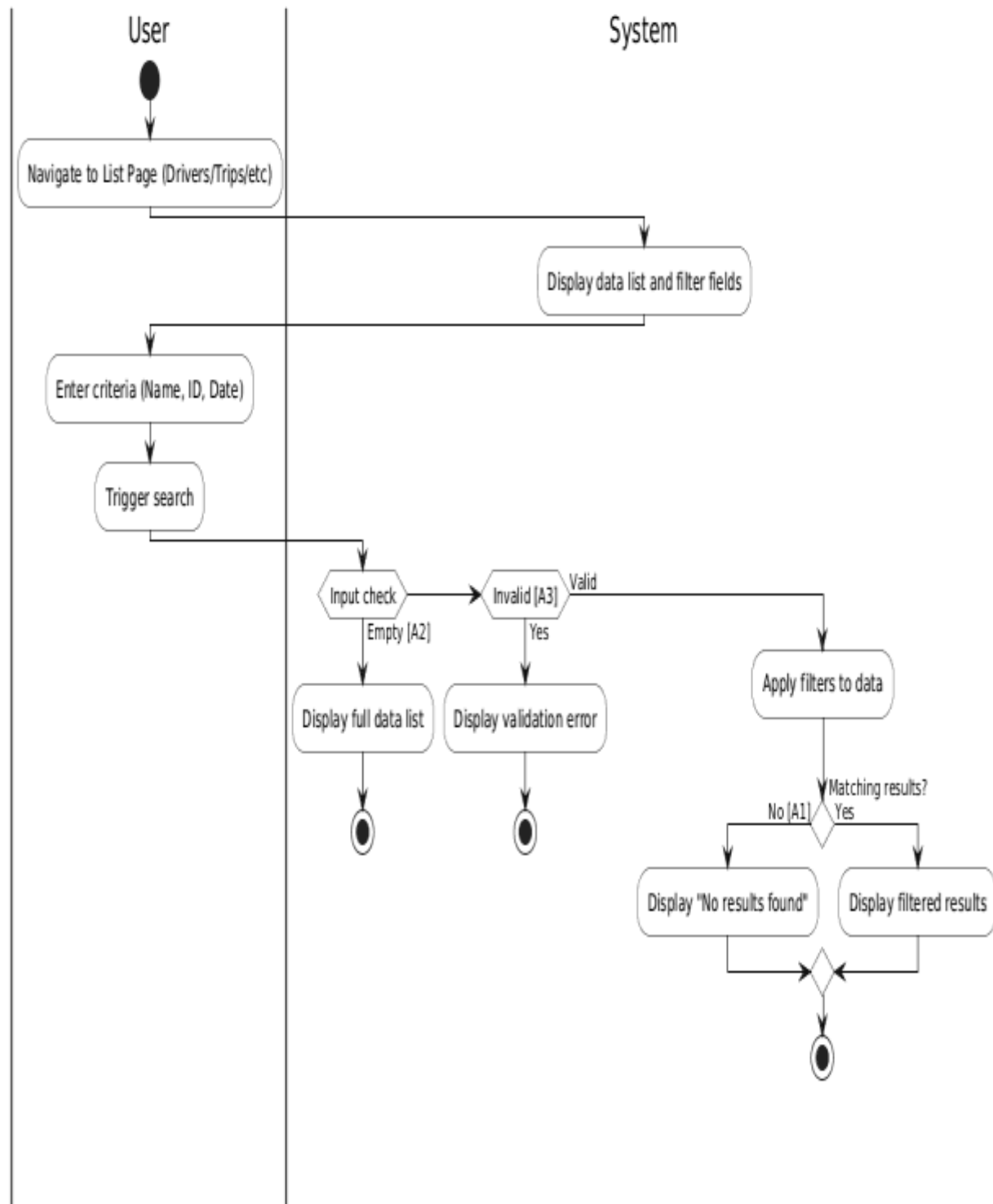


Figure 30 activity UC-21

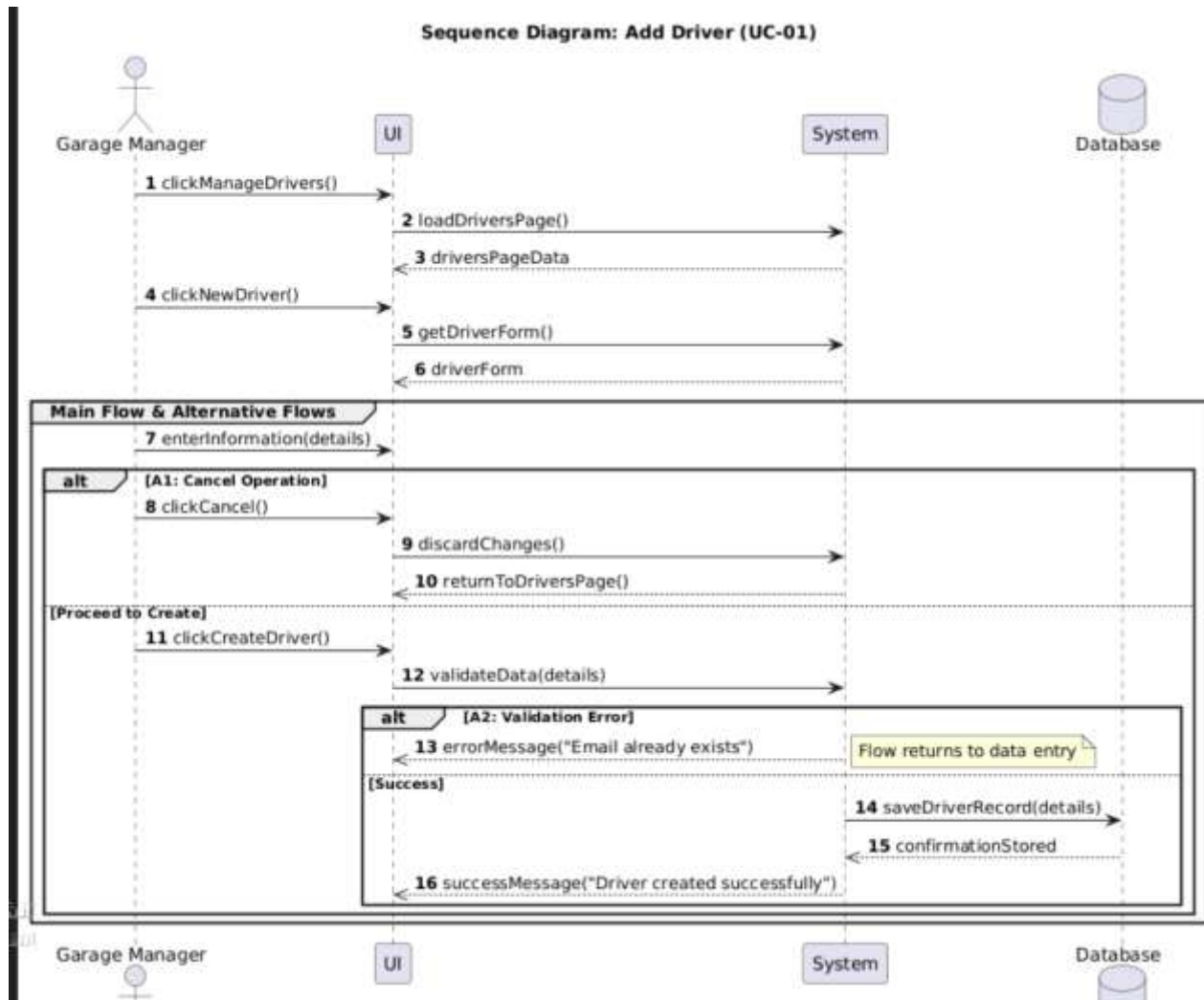


Figure 31 sequence UC-1

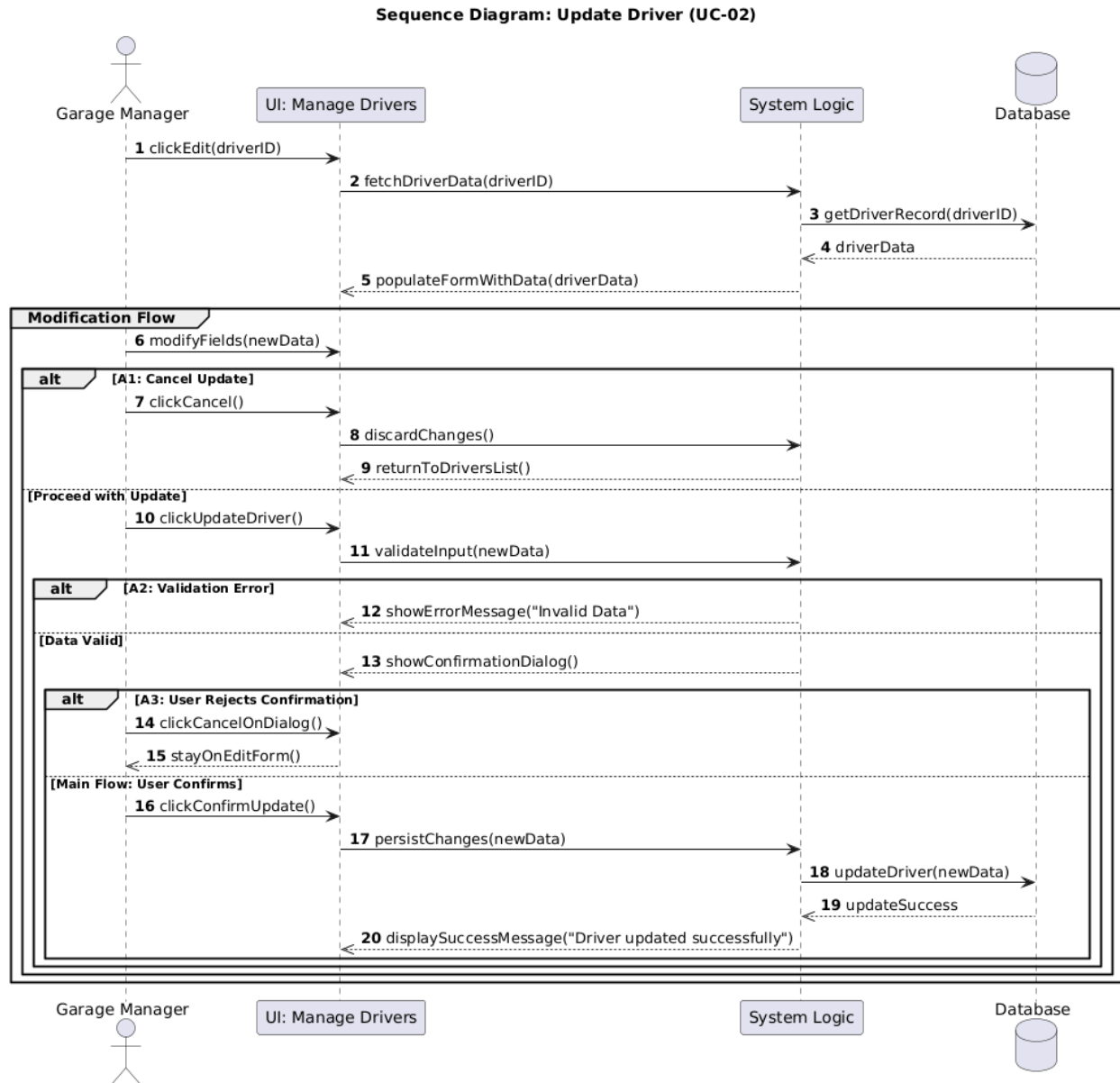


Figure 32 sequence UC-02

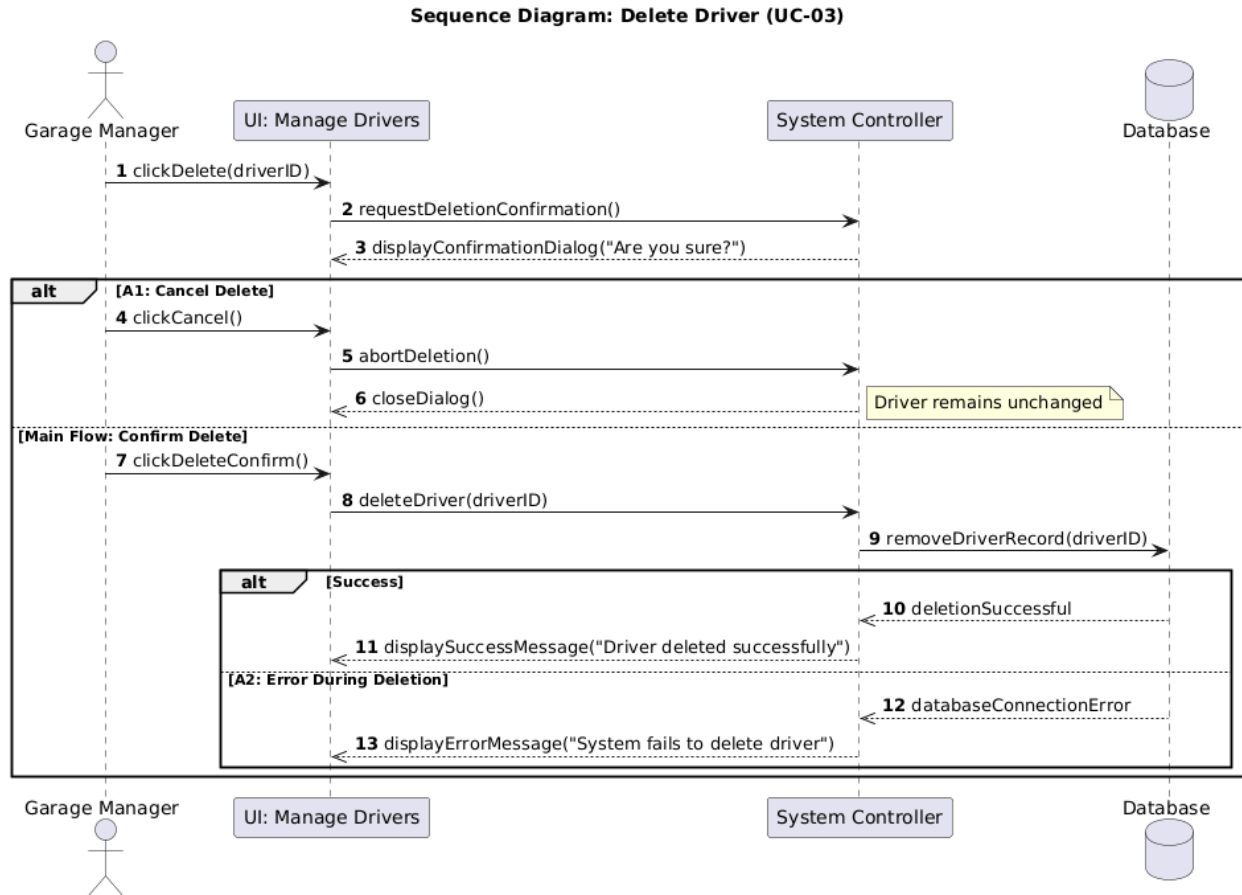


Figure 33 sequence UC-3

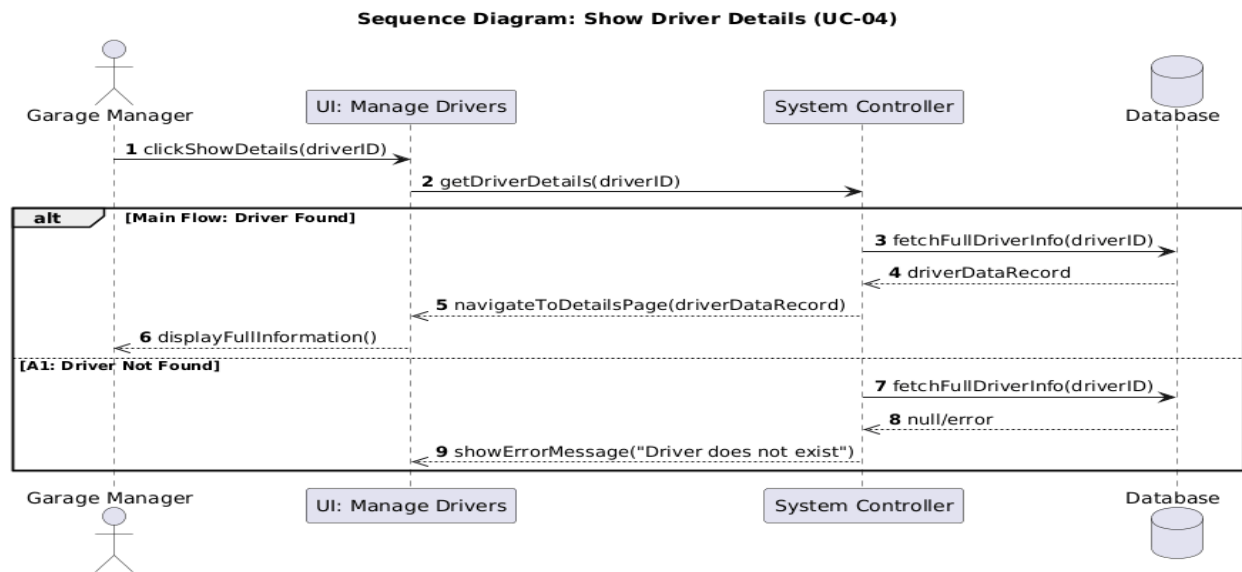


Figure 34 sequence UC-04

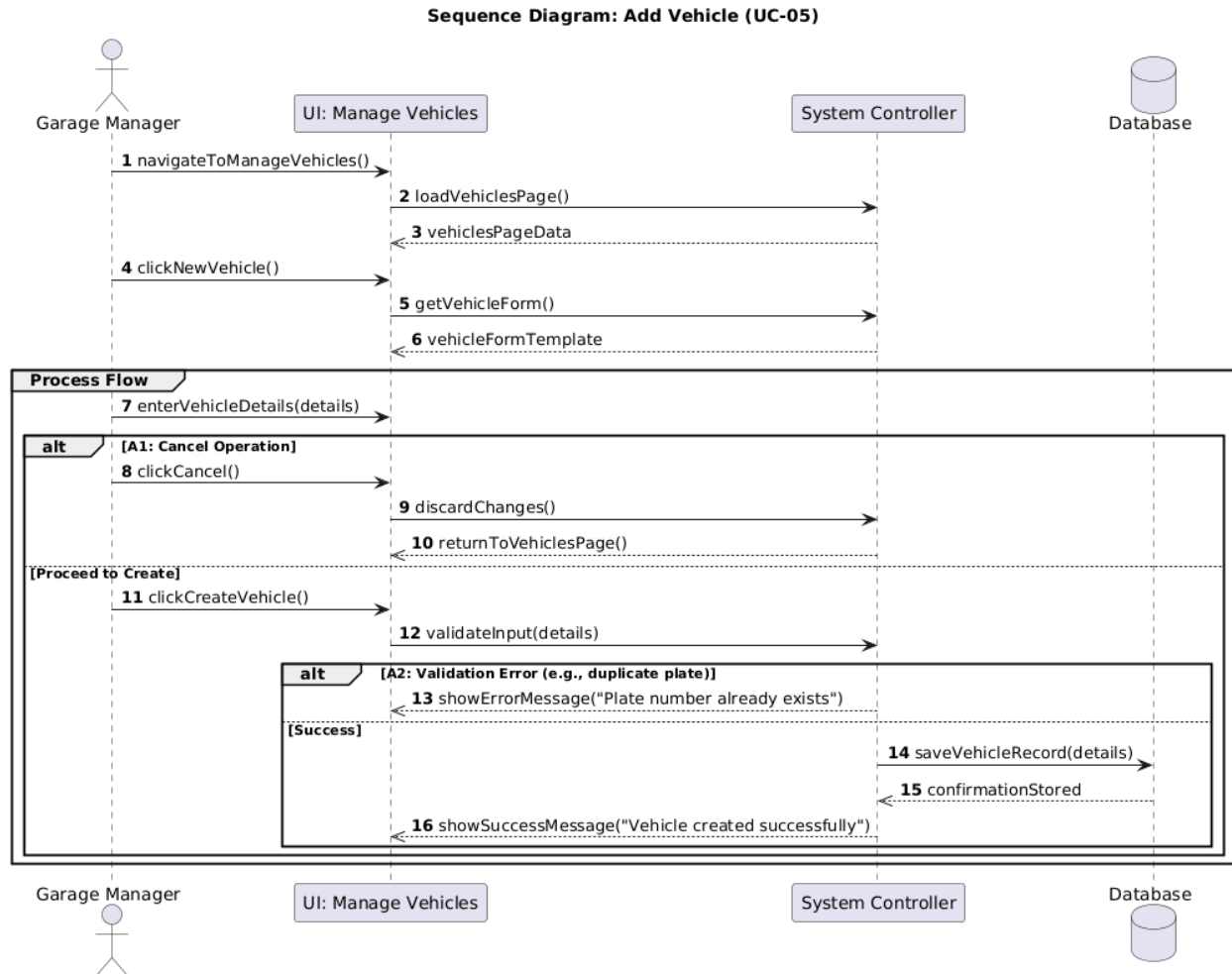


Figure 35 sequence UC-05

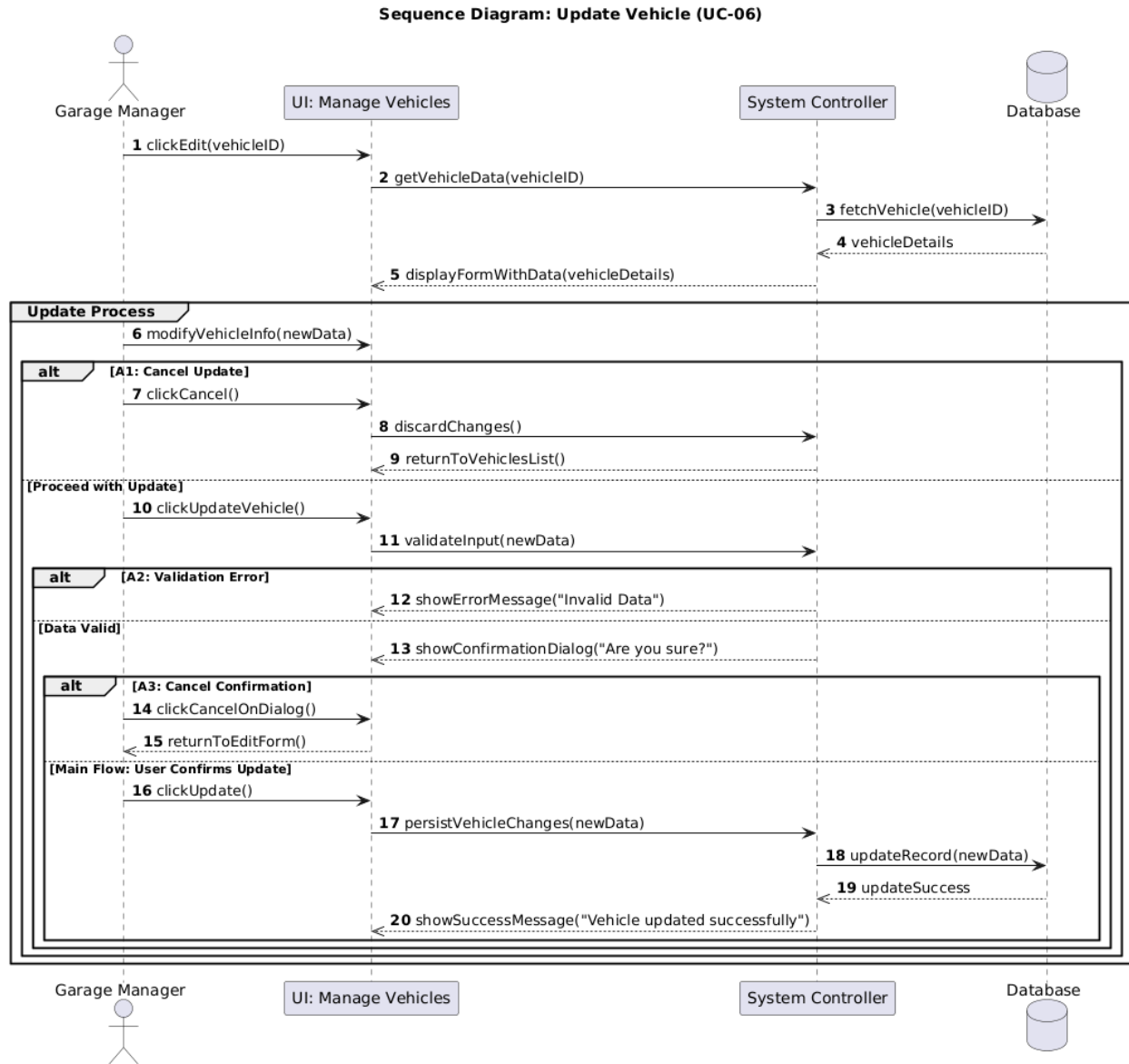


Figure 36 sequence UC-6

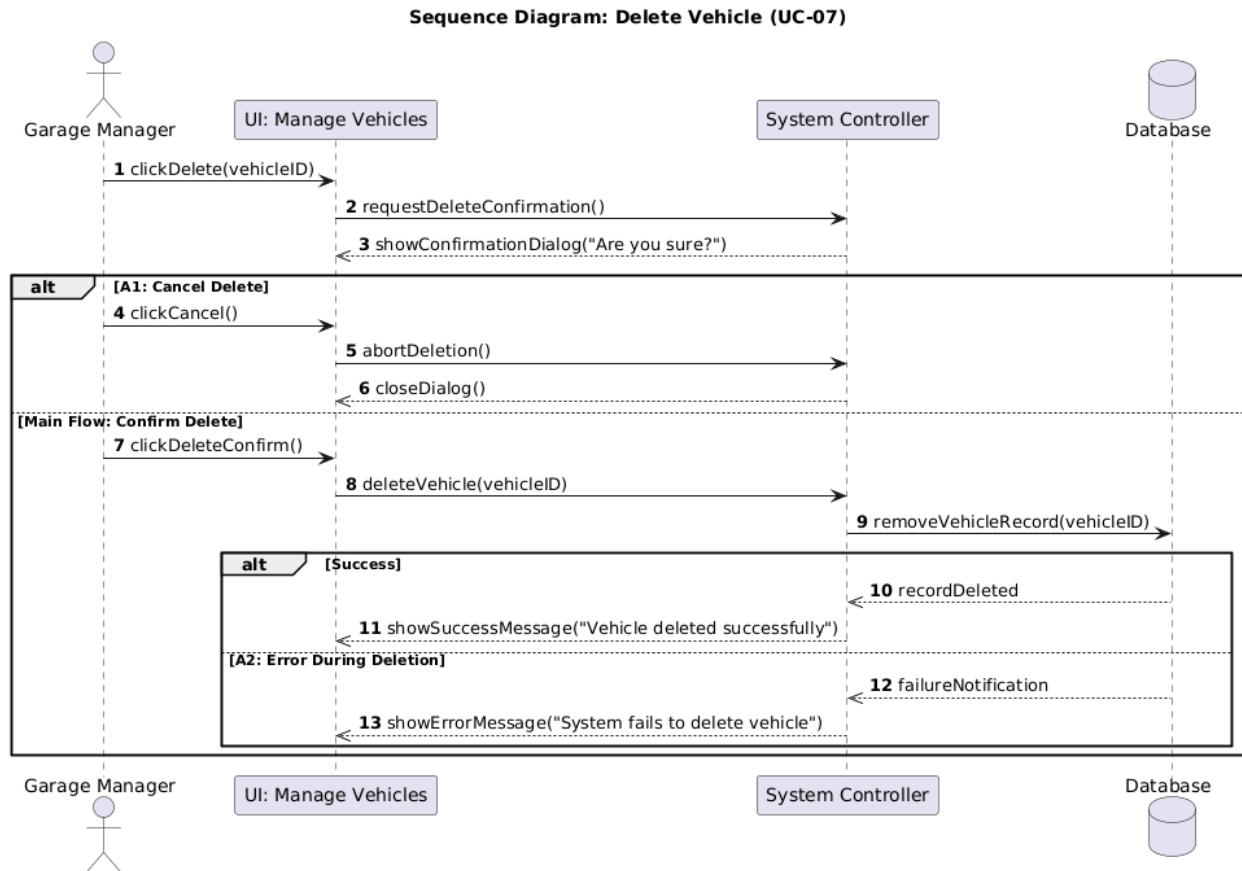


Figure 37 sequence UC-07

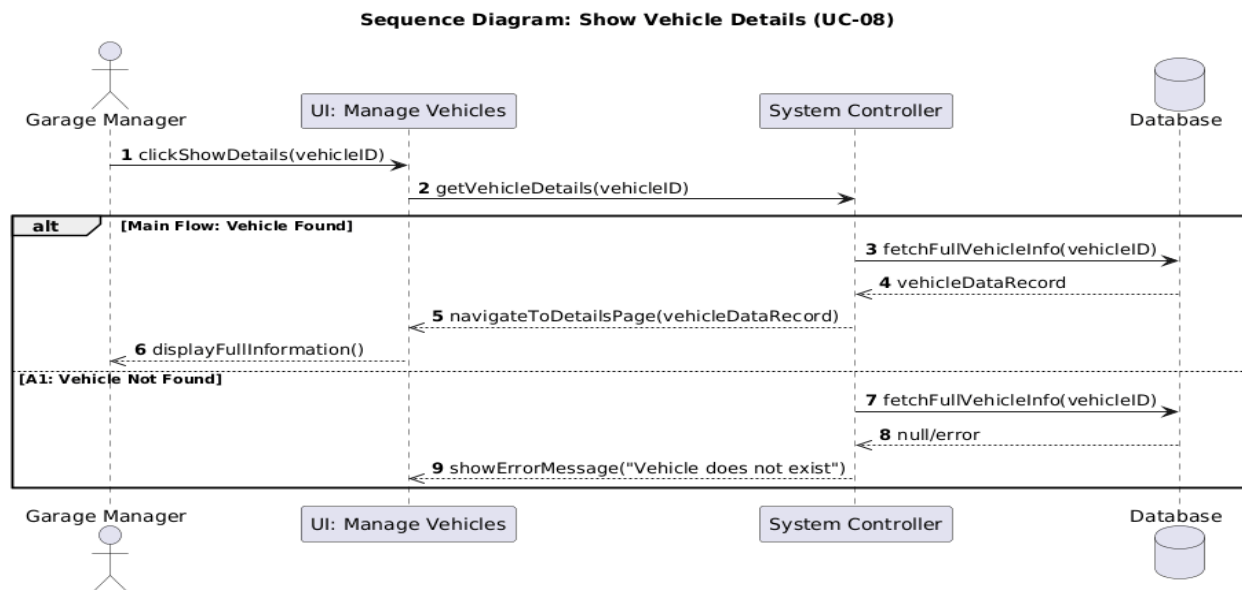


Figure 38 sequence UC-08

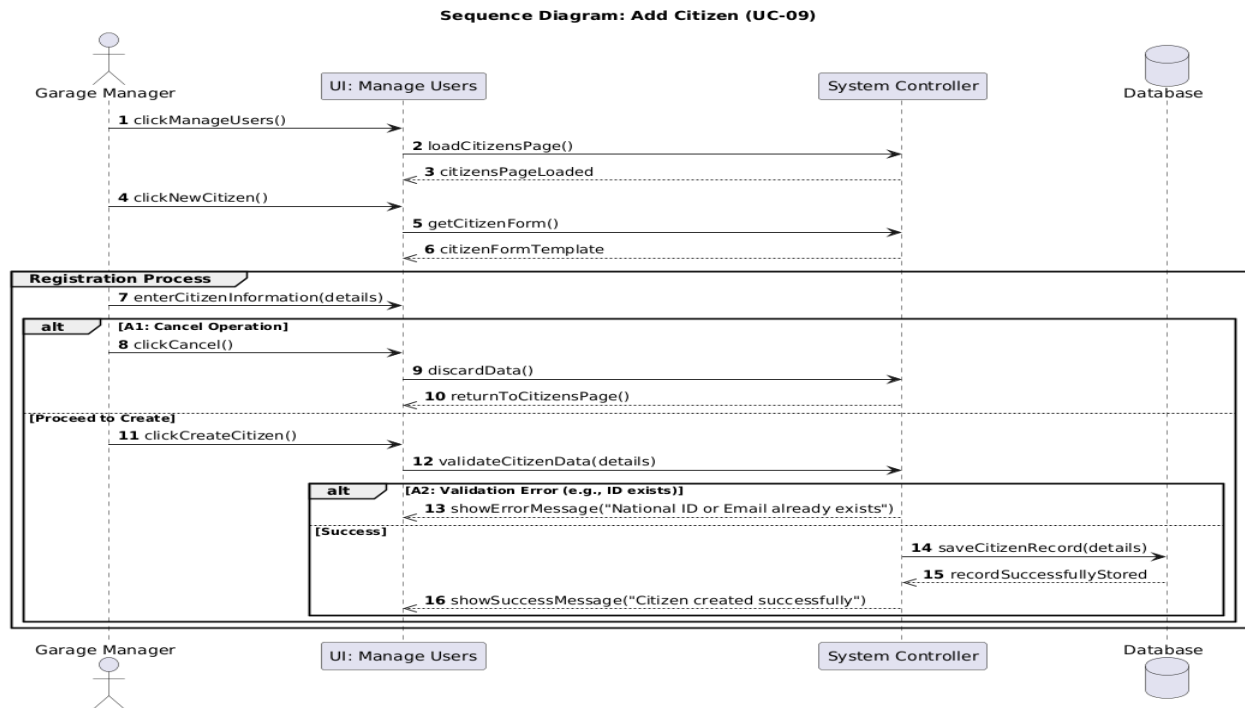


Figure 39 sequence UC-9

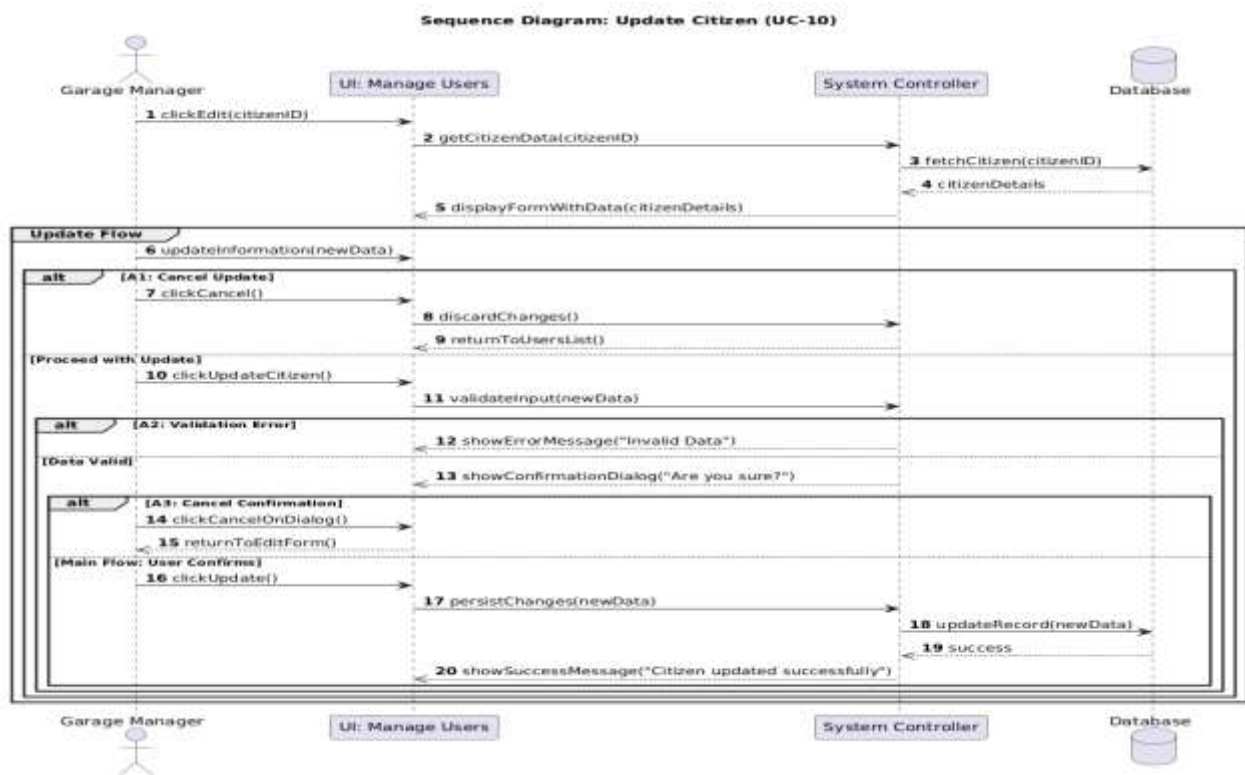


Figure 40 sequence UC-10

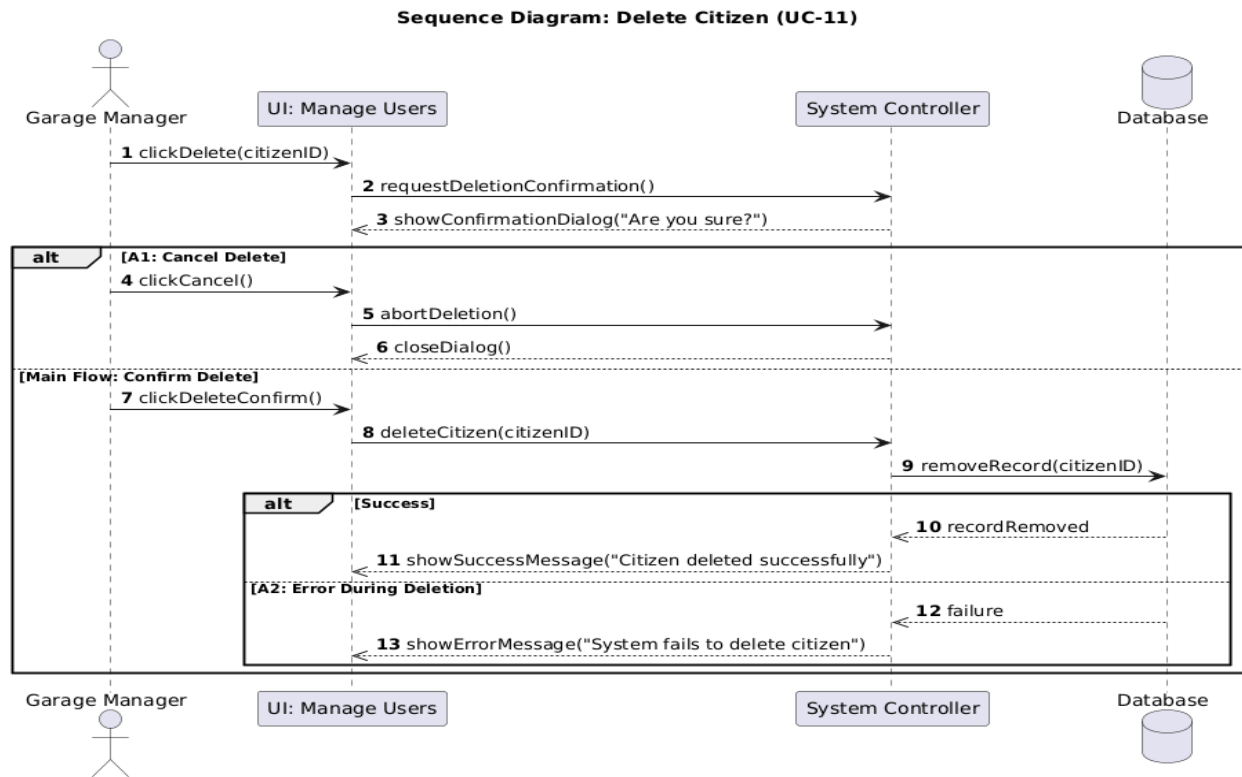


Figure 41 sequence UC-11

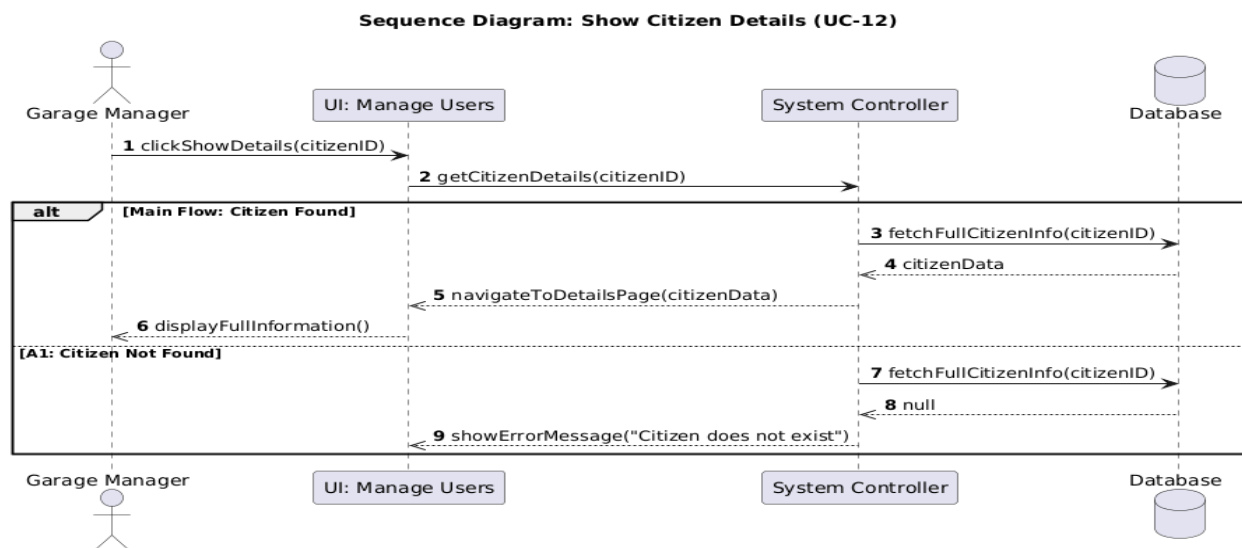


Figure 42 sequence UC-12

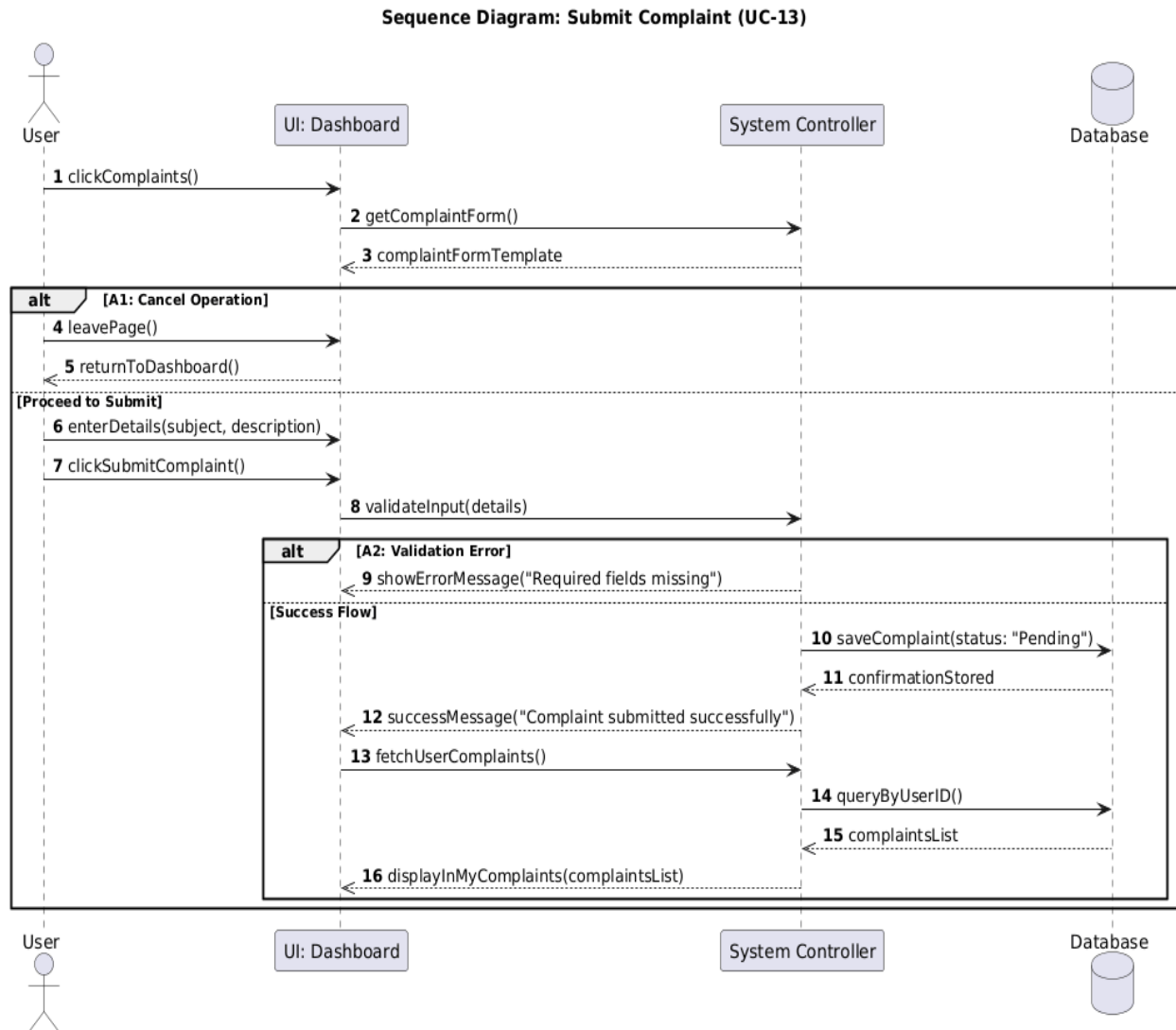


Figure 43 sequence UC-13

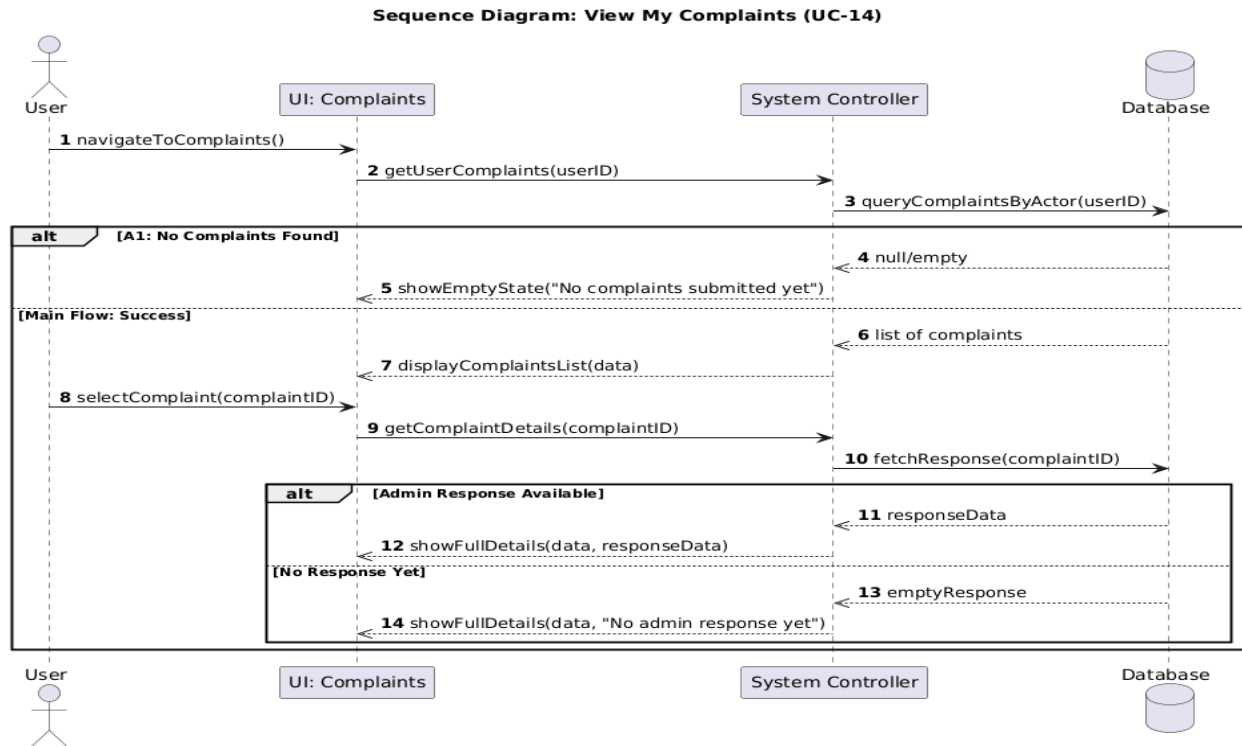


Figure 44 sequence UC-14

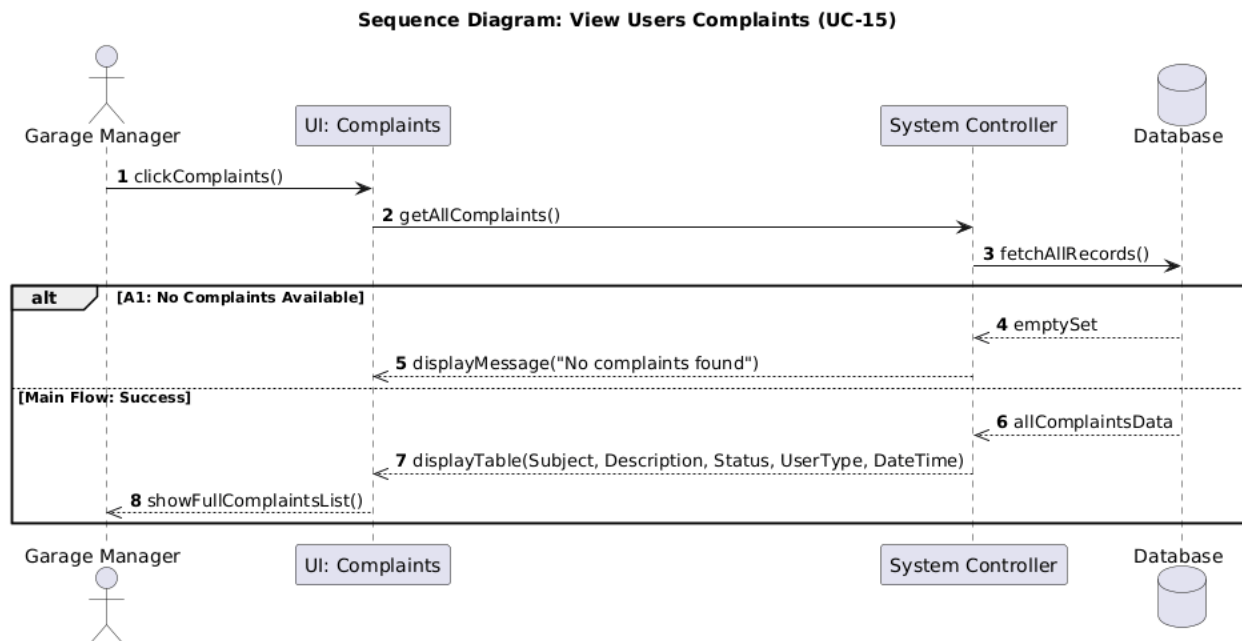


Figure 45 sequence UC-15

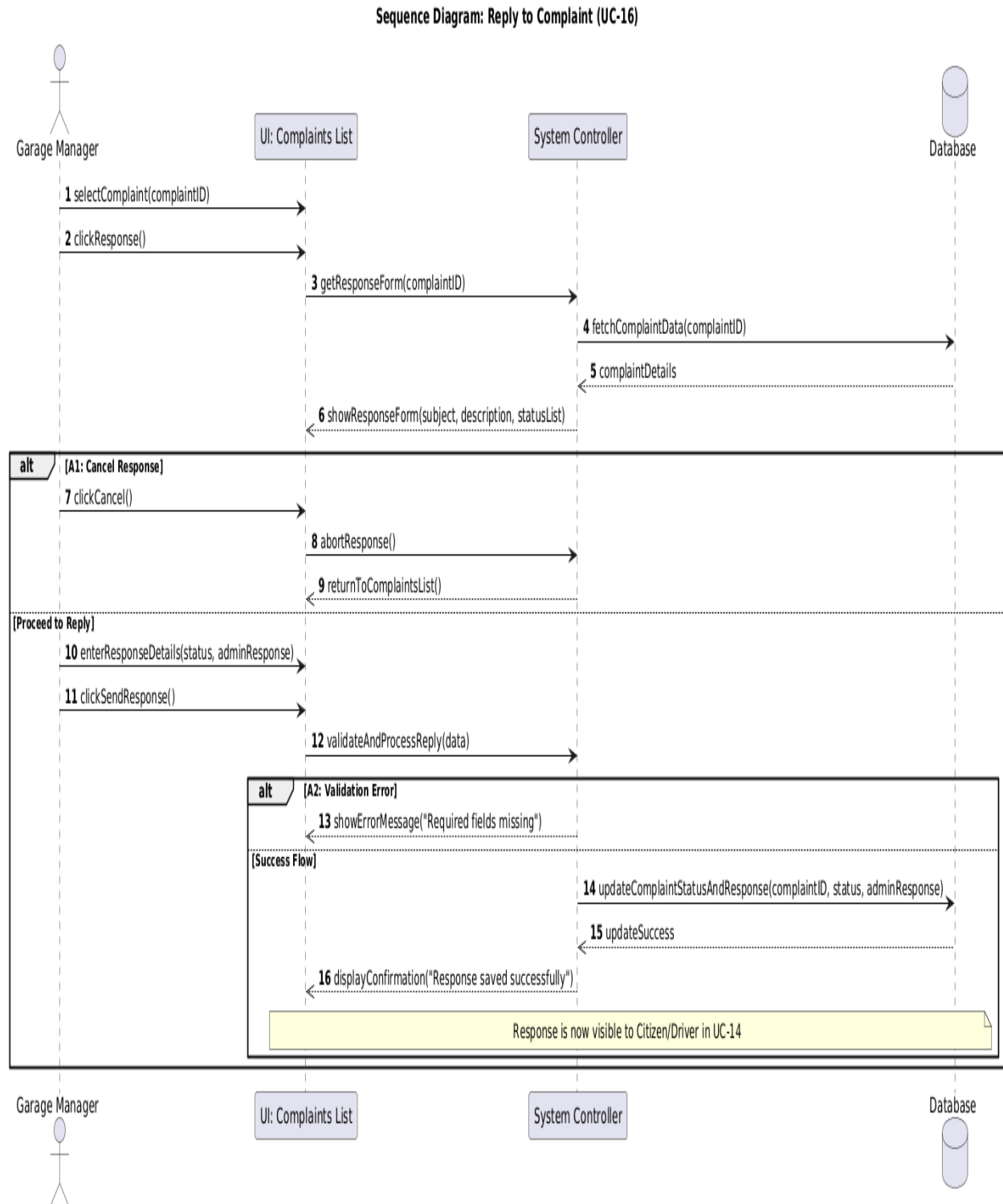


Figure 46 sequence UC-16

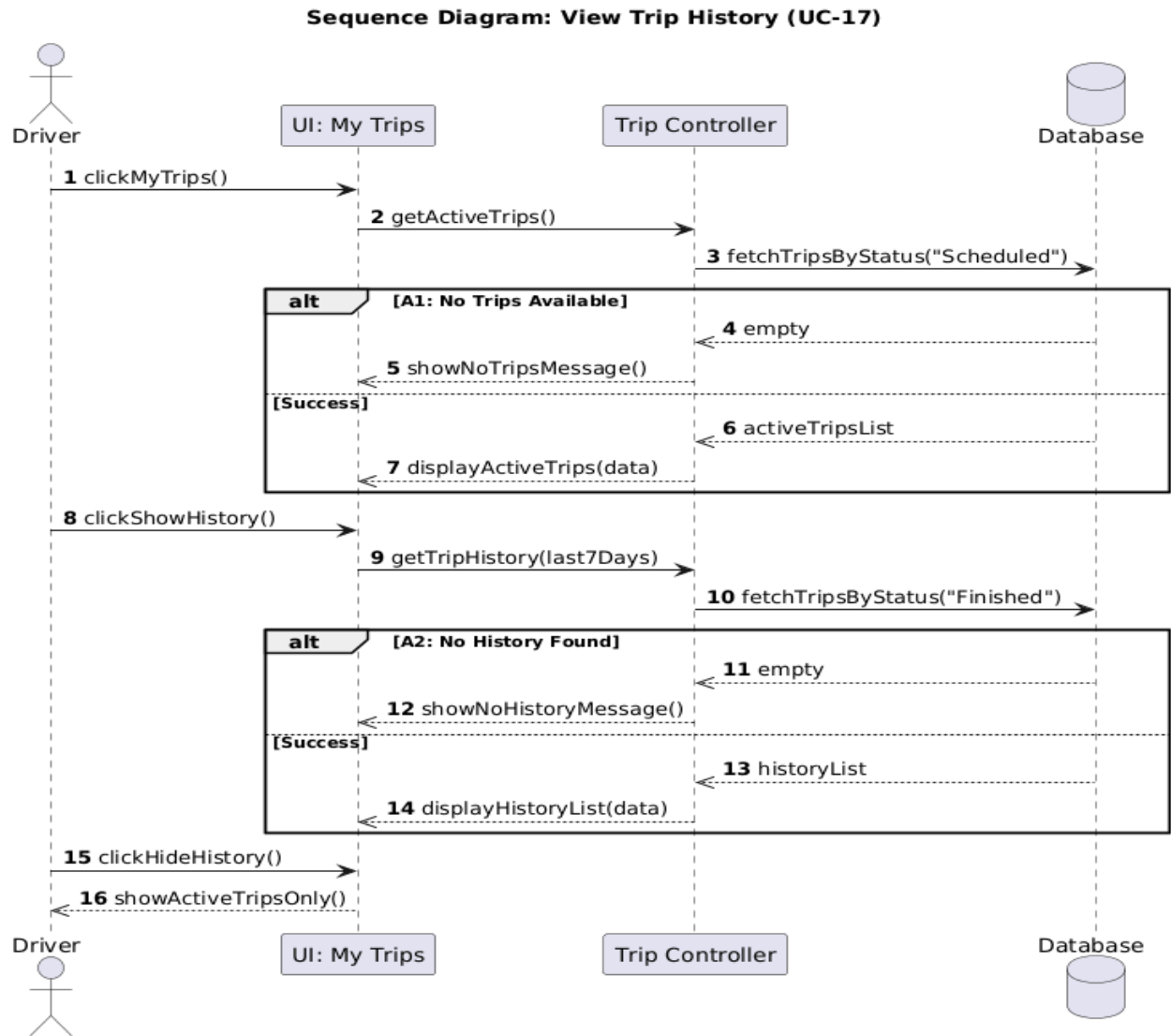


Figure 47 sequence UC-17[

Sequence Diagram: View Booking History (UC-18)

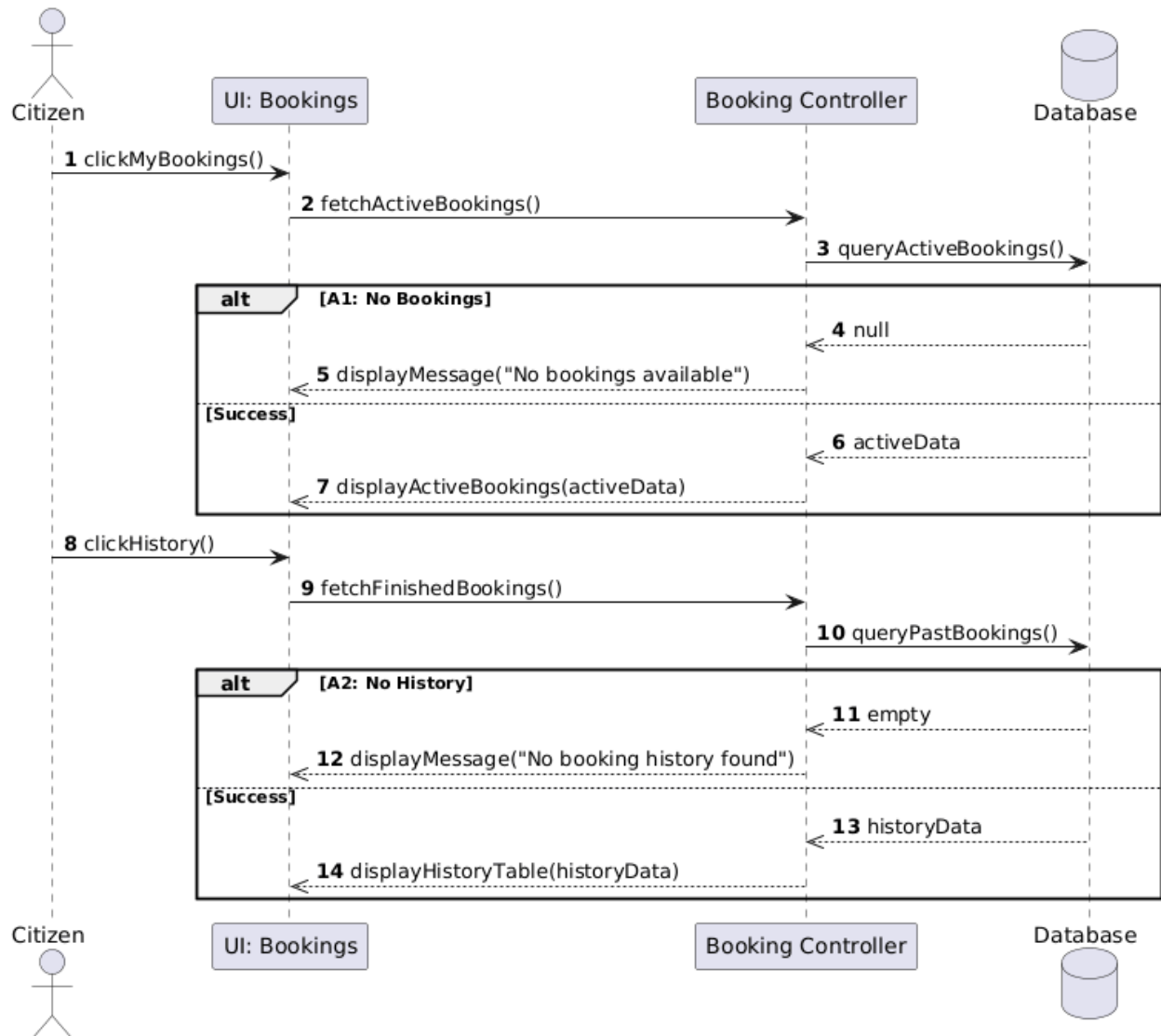


Figure 48 sequence UC-18

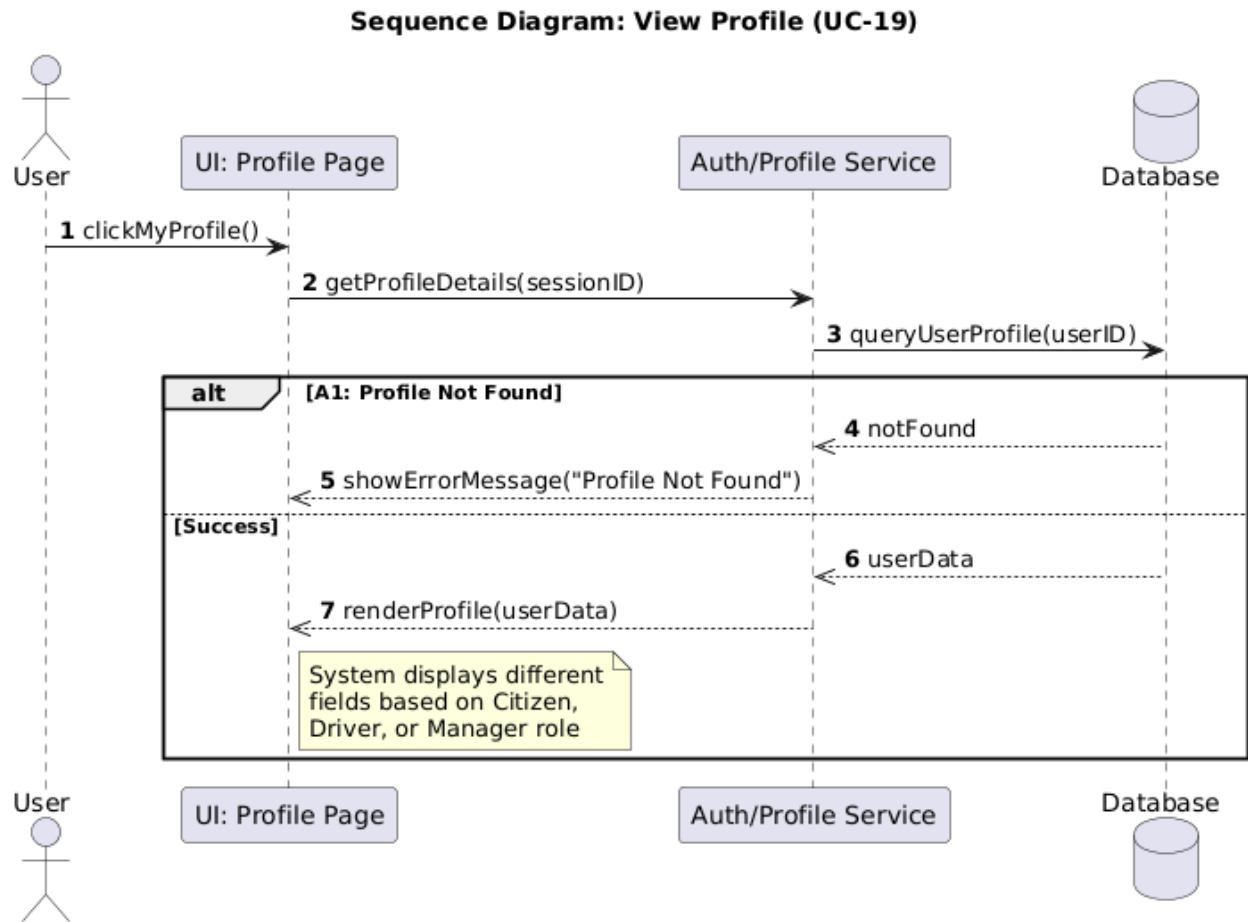


Figure 49 sequence UC-19

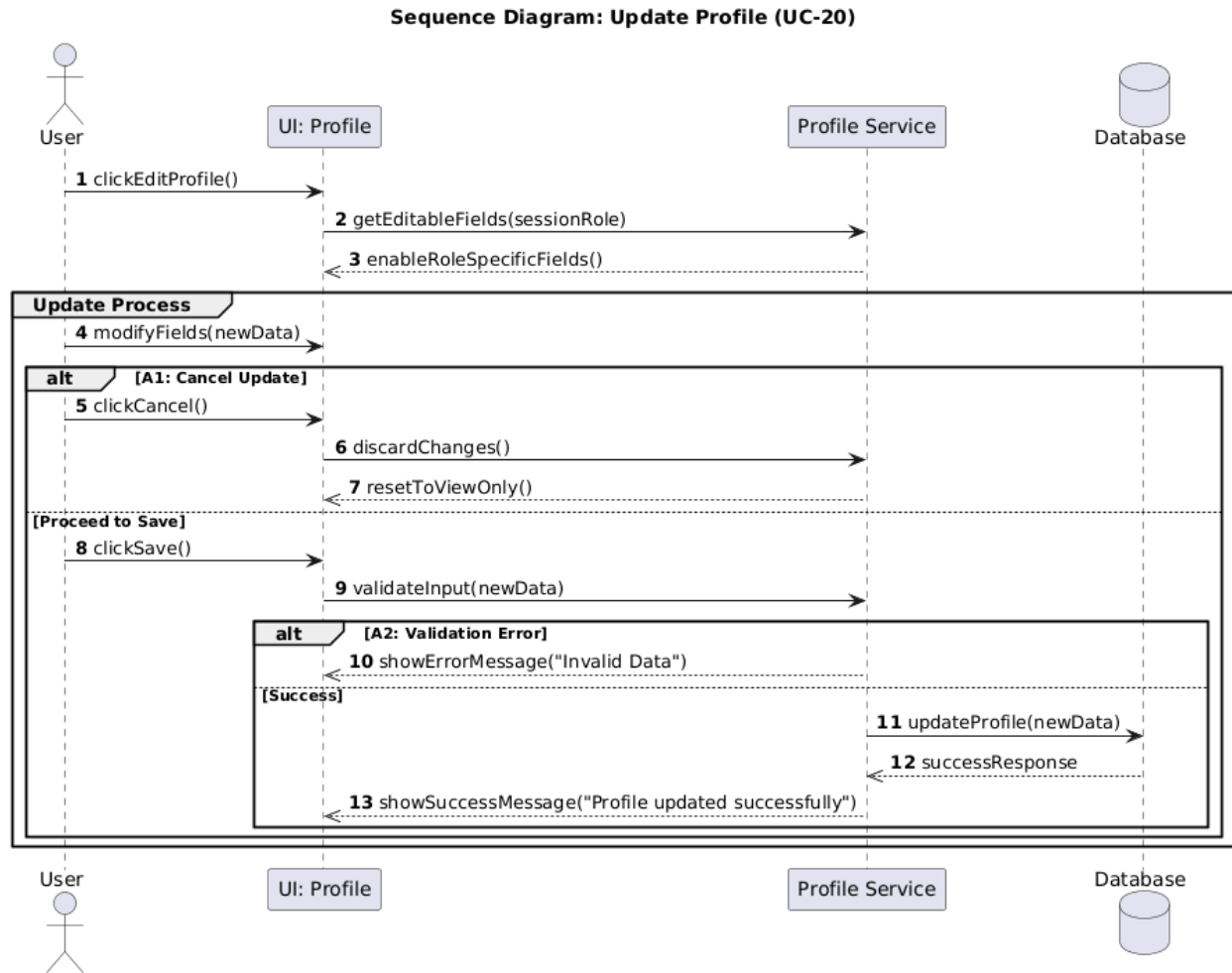


Figure 50 sequence UC-20

Sequence Diagram: Search with Filters (UC-21)

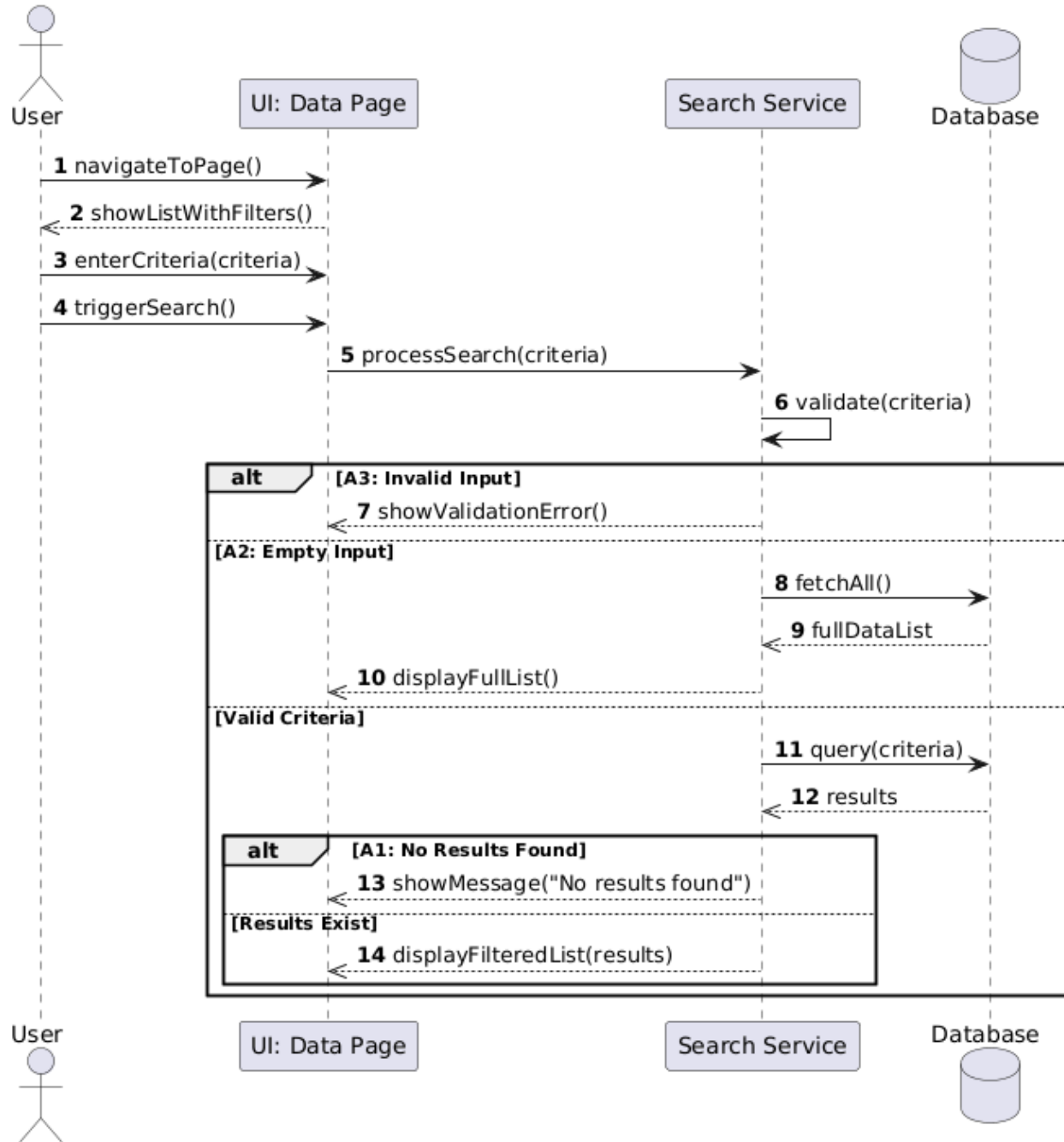


Figure 51 sequence UC-21

7-5-Test Case Descriptions